

The spectrum of a compact internal space. II.

Effective couplings and mass scales

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ABSTRACT: This paper studies the numerical consequences of a fixed compact spectral datum, taken here to be the Laplace and Dirac spectra of RP^3 , together with a specified scale-flow prescription. The structural content supported by this datum, namely the gauge algebra, the fermion representation content, and the positive colour-singlet scalar scale, is recorded in the companion paper, and the present paper evaluates its numerical consequences. The calculation closes the dimensionless window capacity and evaluates a set of effective couplings, mass ratios, and low-energy scales. After the structural datum and the closure prescriptions are fixed, no continuously adjusted phenomenological parameter is introduced in the numerical chain. The resulting quantities are compared with representative Standard Model and cosmological reference values, and the deviations are reported together with the sensitivity to the main closure choices. The deviations are central-value comparisons rather than a global statistical fit. The results are scheme-conditioned consequences of the spectral construction, with core outputs and downstream phenomenological implications distinguished by the tier classification.

KEYWORDS: spectral datum, compact internal space, effective couplings, scale flow, positive scalar mass, numerical evaluation

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1 Introduction

The companion paper [1] develops a spectral approach to gauge structure and to a positive colour-singlet scalar mass. Its basic object is a discrete spectrum with multiplicities, and physical quantities are defined as convergent spectral sums over that spectrum. Its main structural statement is a classification inside a specified admissible closure class: the chiral carrier, compact internal spectrum, anomaly-free determinant condition, minimal non-redundant zero-mode completion, and endpoint spectral-sum closure define the class. Within this class the internal-space geometry, gauge algebra, four-dimensional carrier and its $3 + 1$ reconstruction, fermion representation content, and positive colour-singlet scalar mass are selected. The present paper adopts that spectral datum and evaluates the numerical output of the scale-flow prescription built on it. The resulting values provide the quantitative realisation and observational comparison of the construction.

The gravitational and operator-selection parts of the construction were developed in Refs. [2, 3]. The present paper uses the compact spectral datum and structural closures of the companion paper, and evaluates the resulting fixed point, couplings, mass scales, and cosmological endpoint quantities.

The datum is the Laplace and Dirac spectrum of a compact three-dimensional internal space of constant positive curvature, the real projective space $RP^3 = S^3/\mathbb{Z}_2$. The core inputs of the computation are therefore two-fold. They are this spectrum, inherited from the companion paper, and the closure prescriptions recorded in Section 2, namely the window profile, the pole criterion, the transverse-traceless kernel, and the normalisation

conventions. Changing one of these prescriptions is treated as a counterfactual replacement of the closure, not as a continuous perturbation of a fitted parameter inside the same calculation. The core outputs are the dimensionless window capacity kL that closes the fixed point, the generation count, the effective gauge-coupling normalisations, the primary fermion mass ratios, and the electroweak scale. These are the quantities whose status is most direct within the construction.

Beyond these core outputs, the same chain yields the QCD scale, proton and glueball mass estimates, CKM and CP relations, and cosmological fractions. These form the downstream tests of the extended construction. The baryon asymmetry, dark-energy density, acceleration scale, and primordial helium fraction follow the effective relations stated in their defining sections and are classified as Tier D phenomenological implications.

The construction can be compared with other spectral and algebraic approaches. Connes noncommutative geometry takes a spectral triple as the primitive object and derives the action from the spectral action principle [4–6]. The present construction shares the use of spectral data as carriers of geometric and physical information, but defines physical quantities directly as convergent spectral sums without passing through an action functional, so the two approaches are complementary. The algebraic classification of the finite geometry underlying the Standard Model [7, 8] provides a natural consistency condition, and the heat-kernel expansion of the spectral action [6, 9] is structurally the same object that the spectral sums evaluate, here obtained from the spectrum rather than from the action. Algebraic quantum field theory likewise avoids a path-integral measure [10, 11], but reconstructs observables from nets of local algebras [10, 12], whereas here the algebraic structure is induced from the spectrum. The isomorphism between the spectral-scale flow and the standard renormalisation group equations, recorded in Appendix D.2, provides a bridge between the two [13, 14]. These references locate the construction in relation to existing spectral and algebraic formalisms; the numerical outputs themselves are computed in Sections 5–9 and compared with reference values in Section 10.

The computation is organised around one dimensionful observational anchor, the Newton constant G_N , which converts the dimensionless spectral output into physical units. Every mass scale descends from this anchor through the spectral geometry. The dimensionless chain is fixed by the adopted spectral datum, the window profile, the pole criterion, the normalisation conventions, and the stated downstream effective relations. Appendix D and Section 10.5 quantify the response to counterfactual changes of these defining choices.

The gravitational coupling carries two roles. In its first role G_N is the single observed anchor, and the identity $G_N = 1/(8\pi M_P^2)$ defines the reduced Planck mass. In its second role G_N normalises the long-range transverse-traceless mode selected by the zero-momentum spectral pole of Ref. [2]. Its residue is $Z_{\text{phys}} = \lambda_{\text{long}}/(\lambda_{\text{long}} + \sigma)$. The matter self-energy σ is tiny compared with the long-root spectral eigenvalue λ_{long} , so $Z_{\text{phys}} = 1$ to the displayed precision, and the two roles coincide.

The plan of the paper follows the chain of dependencies among the quantities. Section 2 states the computation scheme, the anchor, and the reproducibility convention. Section 3 closes the window capacity that fixes the internal scale and the generation count. Section 4 records the content symmetries and the torsion parameter. Section 5 computes the gauge

couplings. Section 6 computes the fermion masses. Section 7 computes the electroweak order parameters. Section 8 computes the cosmological quantities. Section 9 computes the QCD scales. Section 10 collects the full parameter table and compares it with observation, closing with a statement of the theoretical sensitivity whose technical details are in Appendix D. The neutrino sector, the CKM and CP content, and the primordial helium abundance are recorded in Appendices E and F as extensions of this core chain. Section 11 discusses the robustness of the central values, the connection to the Standard Model, the acceleration-scale implication, and the principal experimental discriminants.

The scope is fixed at the outset. This paper computes numbers from a fixed spectral datum and inherits the theorem, definition, convention, and assumption classification of the companion paper. Comparison with observed values occurs after completion of the calculation. The window profile, pole criterion, truncation order of the renormalisation-group running, threshold matching, and downstream effective relations are stated explicitly, and Appendix D assesses their consequences.

The derivations that produce the numbers are self-contained at the level of the numerical chain. Appendix C derives the window-capacity fixed point, torsion modulus, coupling conservation law, and mass ladder. The companion paper supplies the structural definitions of the spectral datum and the gauge and fermion content. The present work propagates this fixed completion through the numerical chain, after which the outputs are compared with reference values.

2 The computation scheme

This section fixes the rules of the computation. It states the single dimensionful anchor, recalls the spectral-sum representation, records the closure convention, and specifies how reproducibility is documented.

2.1 The single dimensionful anchor

The only dimensionful quantity taken from observation is the Newton constant G_N . In natural units it is

$$G_N = 6.70883 \times 10^{-39} \text{ GeV}^{-2},$$

the value compiled by the Particle Data Group [15]. From it one forms the reduced Planck mass

$$M_P = \frac{1}{\sqrt{8\pi G_N}} = 2.43532 \times 10^{18} \text{ GeV}.$$

The Newton constant is used only as the observed dimensionful anchor setting the overall mass scale. Every other mass scale in this paper is a dimensionless number multiplying M_P , with the dimensionless number computed from the spectral datum. The full Planck mass $M_{Pl} = \sqrt{8\pi} M_P = 1/\sqrt{G_N}$ appears only where the Hubble parameter or the expansion history needs $1/\sqrt{G_N}$ in the standard normalisation.

The Newton constant G_N converts the dimensionless spectral output into physical units and anchors the gravitational comparison. Dimensionless couplings, mass ratios,

and spectral eigenvalues are computed from the spectral datum and the stated closure conventions; G_N supplies their final dimensional scale.

The inputs of the whole computation fall into three disjoint classes. The first class is the single observed anchor, the Newton constant G_N . The second class is the structural numbers, the pure mathematics 2 , π , $3/2$, and $1/8$, together with the geometric factors $\sqrt{2\pi}$ and $\sqrt{3/\tau}$ recorded where they first appear. The third class is the derived intermediates, the window capacity kL , the torsion modulus τ , the emergence scale M_G , and every later quantity. Each member of the third class is computed from the first two classes by a derivation recorded in this paper, with the full algebra in Appendix C. The chain is therefore fixed from the single anchor and the structural numbers through the stated algebra.

Here “fixed” means fixed by the admissible closure class inherited from the companion paper and by the numerical prescriptions of this section. Replacing the Gaussian window, the pole criterion, or the transverse-traceless normalisation defines a different closure problem.

2.2 The spectral-sum representation

The companion paper defines physical quantities as convergent spectral sums over the spectrum of the internal Laplacian, together with their closed forms where a regularisation is available [1]. This paper uses that representation throughout. The internal spectrum is

$$\lambda_l = \frac{l(l+2)}{L^2}, \quad d_l = (l+1)^2, \quad l = 0, 2, 4, \dots$$

for the scalar Laplacian on RP^3 of radius L , and

$$\pm \frac{n+3/2}{L}, \quad n = 0, 2, 4, \dots$$

for the Dirac operator in the spin structure selected by the chiral content. The positive colour-singlet scalar mass m_δ of the companion paper appears as the spectral lower bound of the colour-singlet scalar sector. Quantities are dimensionless ratios or products of spectral sums, so the dependence on L cancels in the dimensionless output and reappears only through the anchor M_P .

2.3 Internal closure and the structural numbers

The computation is closed in the following sense. A derived quantity is returned from the anchor, the spectrum, and quantities already fixed earlier in the chain. The explicit numerical inputs are the structural numbers 2 , π , $3/2$, and $1/8$, together with the geometric factors $\sqrt{2\pi}$ and $\sqrt{3/\tau}$, whose origins are recorded at their first use. Reference values enter after the computation, in the comparison columns of Section 10 and Appendix A.

2.4 Status of the computed quantities

Four epistemic classes distinguish inherited structure, closure prescriptions, derived outputs, and phenomenological comparisons. Each subsequent section assigns its outputs to one of these classes.

Table 1. The four epistemic classes of the computed quantities. The later sections identify their outputs by class, and no quantity is reported in a class higher than the one it actually occupies.

Class	Meaning and formulation
Structural input	Inherited from the companion paper, namely the internal geometry RP^3 , the gauge algebra, and the fermion representation content. Formulated as <i>assumed input / inherited structure</i> .
Scheme-dependent closure	The Gaussian window, the transverse-traceless kernel, the pole threshold, and the closed value of the window capacity kL . Formulated as <i>prescription / closure choice</i> .
Derived output	Quantities computed within the scheme, namely the gauge couplings, the electroweak scale v , the electron mass m_e , the Hubble rate H_0 , the charged-fermion mass ratios, the absolute neutrino mass scale, and the two oscillation splittings. Formulated as <i>derived quantity within the stated scheme</i> .
Phenomenological comparison	The more remote, model-dependent applications, namely the PMNS and CP relations, the primordial abundances, the dark-energy density, and the acceleration scale. Formulated as <i>comparison / downstream implication</i> .

The four status classes describe the logical role of each quantity in the chain. The separate Tier A–D labels of Section 10 describe the directness of the numerical comparison: fixed data, core outputs, scheme-sensitive quantities, and exploratory downstream extensions, respectively. The Status column of Table 3 and the Tier column of Appendix A record these complementary classifications.

After the structural inputs and closure prescriptions are specified, the numerical chain is fixed. Appendix D quantifies its response to counterfactual changes of the window profile, pole criterion, and normalisation conventions.

2.5 Reproducibility

The derivations form an acyclic chain: each quantity depends only on the Newton anchor, the inherited spectrum, or quantities that precede it. Appendix C records the defining relation for every derived entry, and the parameter table of Section 10 is evaluated from those closed forms. Appendix B records the closure convention in detail.

2.6 The use of renormalisation group equations

In this paper, the standard two-loop RG equations are used to evolve couplings from the emergence scale M_G down to the Z -boson mass M_Z . This evolution occurs in the regime

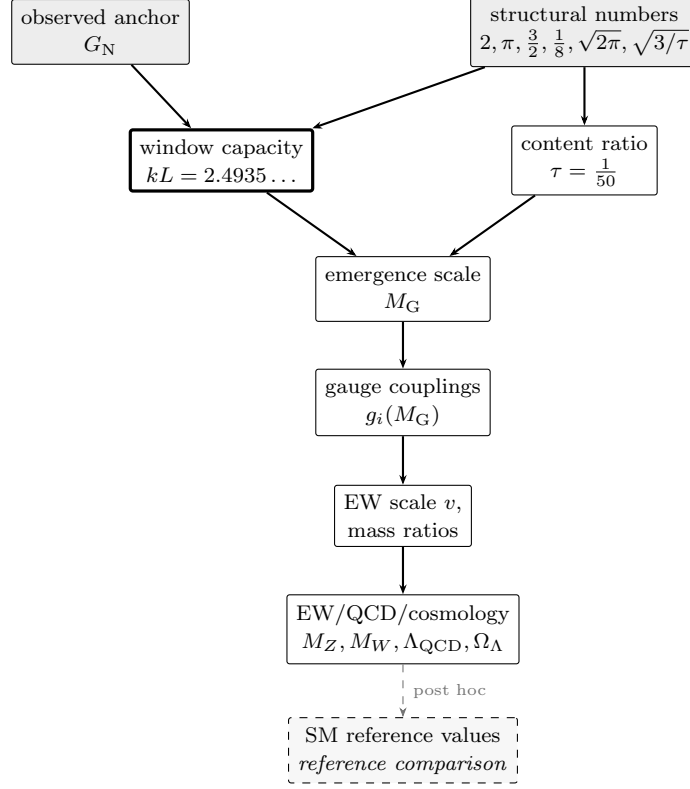


Figure 1. Single-anchor dependency graph. The computation is a directed dependency graph rooted in one observed anchor G_N and the structural numbers; every downstream quantity is a closed-form function of these and of the self-consistent window capacity kL (the central node, highlighted). The Standard-Model reference values (dashed) are used only after the internal quantities have been computed.

where the spectral sums are dominated by the zero modes and the lowest Kaluza–Klein excitations, the same regime in which the spectral-sum definition and the standard measure-theoretic formulation coincide, as shown in Appendix C of the companion paper [1]. The RG equations are therefore used as the low-energy expression of the spectral-scale flow in the shared effective regime. The interacting corrections above the emergence scale are handled by the spectral sums themselves. Below that scale, the standard perturbative running is a controlled approximation whose accuracy is recorded in Section 10.4. The precise sense in which the spectral-scale flow matches the RG equations is recorded in Appendix D.2.

The dependency structure of the whole computation is summarised in Figure 1.

3 The internal scale and the window capacity

This section fixes the internal scale of the construction and derives the generation count from it. It is the anchor of the computation, in the sense that every later section reads its numbers from here. The section records the spectral datum, the Kaluza–Klein descent

from the Planck anchor, the closure of the window capacity, and the resulting generation count.

3.1 Scope and sensitivity of the closure

The window capacity kL is the pivotal intermediate of the computation: the emergence scale, gauge couplings, fermion masses, and electroweak scale all descend from it. It is fixed by the Gaussian window profile, the transverse-traceless response, and the pole normalisation $4/27$ recorded in Section 2. The resulting dimensionful hierarchy is exponentially sensitive to kL . Since $v \propto e^{-4\pi kL}$ and $m_e \propto e^{-20kL}$, a one-percent counterfactual displacement of kL moves v by -32% , m_e by -50% , and ρ_Λ by -262% , while the content ratio Ω_Λ has zero elasticity. Appendix D records the full elasticity matrix and representative responses around the adopted closure.

3.2 The spectral datum

The internal space is $RP^3 = S^3/\mathbb{Z}_2$, the real projective space of radius L . Its relevant spectral data are fixed in the companion paper [1] and collected here. The scalar Laplacian has the spectrum

$$\lambda_l = \frac{l(l+2)}{L^2}, \quad d_l = (l+1)^2, \quad l = 0, 2, 4, \dots,$$

where the restriction to even l is the antipodal projection, and d_l is the multiplicity. The Dirac operator in the spin structure selected by the chiral content has the spectrum

$$\pm \frac{n+3/2}{L}, \quad n = 0, 2, 4, \dots,$$

again restricted to the even sector of the tower. The volume is $\text{Vol}(RP^3) = \pi^2 L^3$, half the round-sphere volume. The number of Killing vectors is six, the dimension of the isometry algebra $\mathfrak{so}(4)$. The first cohomology vanishes, $H^1(RP^3, \mathbb{R}) = 0$, so there is no harmonic one-form, and the scalar sector has no zero mode beyond the constants.

3.3 Kaluza-Klein descent from the Planck anchor

The spectral datum has one free scale, the radius L . This scale is fixed by the descent from the Planck anchor. The companion paper records the product-space construction in which the internal spectrum is lifted to four dimensions and the heavy modes decouple, leaving a four-dimensional effective theory whose mass scales are the internal eigenvalues measured in Planck units [16, 17]. The descent chain is

$$G_N \rightarrow M_P = \frac{1}{\sqrt{8\pi G_N}} \rightarrow M_G = \frac{M_P \sqrt{\pi}}{kL},$$

where M_G is the emergence scale and kL is the dimensionless window capacity introduced below. The Kaluza-Klein masses are

$$m_n = \frac{n+3/2}{kL} M_G,$$

and the window edge is

$$kL M_G = M_P \sqrt{\pi}.$$

The spectral-pole closure of the next subsection fixes the remaining dimensionless quantity kL .

3.4 Closure of the window capacity

The window capacity kL is the self-consistent fixed point of the spin-2 spectral-pole condition. The defining object is the transverse-traceless two-point amplitude at zero external momentum, evaluated as a spectral sum over the discrete RP^3 spectrum,

$$\Pi_{\mu\nu}^2 = \frac{1}{V_3} \sum_{\text{fields}} \sum_{\text{modes}} d_n w \int_0^\infty \frac{d\omega}{\pi} \frac{(\omega^2 + \lambda_n)^2}{(\omega^2 + \lambda_n + m_{\text{eff}}^2)^2} e^{-(\omega^2 + \lambda_n)/\Lambda^2},$$

with $V_3 = \pi^2 L^3$ the internal volume, d_n the mode multiplicity, and the weights $w = 4$ for each gauge polarisation, $w = 4$ for each Weyl fermion, and $w = 1$ for each scalar. The positive spectral kernel satisfies $K_{\text{TT}}(k^2, 0) = 1$. The Ward-normalised flat-space reference is carried by the corresponding subtracted amplitude, while the curvature and torsion shifts determine the RP^3 spectral response. The full definition is recorded in Appendix C.1.

The massless-pole threshold is the normalisation extremum

$$\frac{4}{27} = \max_{0 \leq y \leq 1} y(1-y)^2 \quad \text{at} \quad y = \frac{1}{3},$$

the normalisation of the spin-2 spectral-pole condition, where $y = m^2/(k^2 + m^2)$ is the mass fraction of the transverse-traceless kernel $k^4/(k^2 + m^2)^2 = (1-y)^2$. The emergence scale is the fixed point at which the long-range transverse-traceless response develops its zero-momentum pole,

$$\frac{V_3 \Pi_{\mu\nu}^2(kL, k^2/M_P^2)}{32\pi^2} = \frac{4}{27} \quad \text{at} \quad k = M_G.$$

On the self-similar flow $L(k) = C/k$ with $C = M_P \sqrt{\pi}$ this is one equation in one unknown. It is solved by bisection on the interval $[0.3 M_G, 3 M_G]$ with the stopping width 10^{-15} , and the self-consistency between $M_G = C/kL$ and the pole scale is iterated to the displayed precision. The fixed point is

$$kL = 2.49353.$$

Proposition 3.1 (Unique window-capacity fixed point). *On the self-similar flow $L(k) = C/k$ with $C = M_P \sqrt{\pi}$, the fixed-point function*

$$F(kL) = \frac{V_3 \Pi_{\mu\nu}^2(kL, k^2/M_P^2)}{32\pi^2} - \frac{4}{27}, \quad k = M_P \sqrt{\pi}/kL,$$

is monotonic on the interval $kL \in [0.3, 3]$ in the numerical evaluation of the spectral sum, and changes sign once. The spectral-pole closure $F(kL) = 0$ therefore has a unique solution, to which the bisection with the self-consistency iteration converges.

Proposition 3.2 (Weak-coupling comparison of the window capacity). *The window capacity is determined by the spectral-pole closure $F(kL) = 0$ (Proposition 3.1). After that closure has been made, the window-capacity form $\alpha_W = 2/kL^5$ can be inverted using the reference weak coupling; the resulting post hoc comparison value of kL agrees with the spectral-pole value to within 0.0002%.*

The spin-2 spectral-pole condition fixes kL . The subsequent weak-coupling inversion is its post-computation cross-check against the Standard Model reference value. Appendix B records this dependency order. The Gaussian width of the scale field is $L_{\text{Cg}} = \sqrt{\pi}$, which fixes the critical curvature

$$R_c = \frac{6}{\pi}.$$

The window capacity is then

$$(kL)^2 = 6.2177,$$

which sits below the Euclidean-period value $2\pi = 6.2832$ by 1.04%.

3.5 The generation count

The number of fermion generations is the number of spinor modes of the internal tower that fit inside the coarse-graining window. The criterion is

$$n + \frac{3}{2} < (kL)^2.$$

With $(kL)^2 = 6.2177$ the retained modes are

$$n = 0 \text{ (1.5)}, \quad n = 2 \text{ (3.5)}, \quad n = 4 \text{ (5.5)},$$

while $n = 6$ gives 7.5, which lies above the edge and is excluded. The window therefore contains exactly three modes, and the generation count is

$$n_g = 3.$$

The mode mass ladder is $m_n \propto e^{-\alpha n}$ with the extrusion order $n = \{0, 2, 4\}$, so the three generations are ordered by their extrusion order along the scale flow, and the hierarchy among them is the geometric hierarchy recorded in Section 6. This closes the two quantities that the companion paper left open, the numerical value of the window capacity and the generation count that follows from it. The step structure of the criterion is shown in Figure 2.

4 Content symmetries and the torsion parameter

This section records the counting symmetries of the fermion content and derives the torsion parameter from them. The content is fixed by the companion paper, which obtains the five representations and the hypercharge assignment from anomaly cancellation and the Yukawa structure. The present section turns that content into the numerical identities that the later sections read.

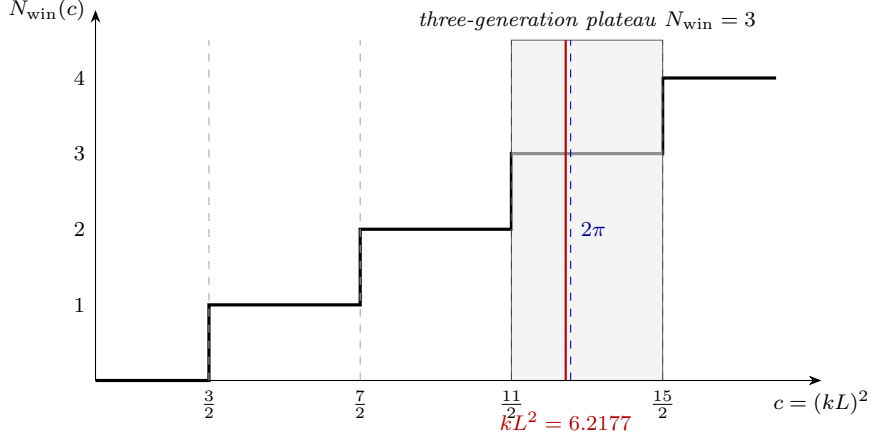


Figure 2. Window-capacity plateau. The generation count $N_{\text{win}}(c)$ is a step function of the window capacity $c = (kL)^2$, with thresholds at $n + \frac{3}{2}$ for $n = 0, 2, 4, 6$. The value $N_{\text{win}} = 3$ is a plateau over the whole interval $\frac{11}{2} < c < \frac{15}{2}$, so the count is insensitive to the precise value of c . Both the self-consistent closure value $kL^2 = 6.2177$ and the Gaussian benchmark $2\pi = 6.2832$ lie inside this plateau.

4.1 Content counting

The five representations per generation are $Q_L = (3, 2)$, $u_R = (3, 1)$, $d_R = (3, 1)$, $L_L = (1, 2)$, and $e_R = (1, 1)$. The left-handed components are the two doublets, Q_L with $3 \times 2 = 6$ components and L_L with $1 \times 2 = 2$ components, so

$$N_L = 6 + 2 = 8.$$

The right-handed singlets give

$$N_R = 3 + 3 + 1 = 7.$$

The number of Weyl fermions per generation is

$$N_f = N_L + N_R = 15,$$

and the total over three generations is 45. The left-handed count equals the number of colour generators

$$N_L = N_g = N_c^2 - 1 = 8,$$

and the right-handed count is one less. This is the content-gauge symmetry recorded and proved in the companion paper. The chiral asymmetry is

$$N_L - N_R = 1,$$

an odd integer, and its parity is the non-trivial element of $\pi_1(RP^3) = \mathbb{Z}_2$, the non-trivial spin structure of the internal space.

The hypercharge assignment satisfies

$$\sum Y = 0,$$

the vanishing of the mixed gravitational anomaly, and its first non-zero moment is

$$\sum Y^2 = \frac{10}{3},$$

the sum $6 \cdot (1/6)^2 + 3 \cdot (2/3)^2 + 3 \cdot (1/3)^2 + 2 \cdot (1/2)^2 + 1 \cdot (1)^2$. The fermion content also obeys the identity

$$N_f = 2 \sum Y^2 \Delta_f^2 = 15,$$

with the fermion conformal weight $\Delta_f = 3/2$, a relation between the Weyl count, the hypercharge moment, and the conformal weight that the torsion parameter of Section 4.3 uses.

4.2 Closure of the conformal-gauge duality

The companion paper fixes the colour algebra by the minimality principle, its third application after $SU(2)$ over $U(2)$ and RP^3 over the lens spaces, and derives the conformal-gauge duality as an identity of those selections. The present section verifies its numerical content. The conformal coupling of a scalar in dimension d is

$$\xi = \frac{d-2}{4(d-1)},$$

and with $d = 3$ it is $\xi = 1/8$. The duality states

$$N_g \xi = 1,$$

equivalently $N_g \Delta = 2(d-1)$ with the scalar conformal weight $\Delta = (d-2)/2 = 1/2$. Substituting $\xi = 1/8$ gives $N_g = 8$, and $N_c^2 - 1 = 8$ has the single positive solution $N_c = 3$. The duality is therefore the statement that the conformal weight per generator is the conserved unit, and it ties the spacetime dimension to the colour number through $d = N_c = 3$.

4.3 The torsion parameter

The totally antisymmetric torsion is parametrised by

$$T^a{}_{bc} = \frac{\tau}{L} \varepsilon^a{}_{bc},$$

where the dimensionless modulus τ is the normalised chiral-content invariant. Its numerator is the chiral excess per fermion component,

$$\langle \chi \rangle = \frac{N_L - N_R}{N_f} = \frac{1}{15},$$

and its denominator is the anomaly-normalised quadratic hypercharge moment,

$$\Pi_{\text{ren}} = \sum Y^2 = \frac{10}{3}.$$

Thus

$$\tau = \frac{\langle \chi \rangle}{\Pi_{\text{ren}}} = \frac{N_L - N_R}{N_f \sum Y^2} = \frac{1}{15 \cdot 10/3} = \frac{1}{50}.$$

The $\mathbb{Z}_2$ spin structure supplies the parity class of the numerator, and anomaly cancellation fixes the hypercharge normalisation in the denominator. Appendix C.2 records the same content ratio and its subsequent appearances in the spectral shifts and squash response.

4.4 Geometric dynamics

The torsion enters the spectrum and the effective potential through the geometric quantities that the later sections use. The $J = 2$ Lichnerowicz eigenvalue of the Einstein-Cartan sector is

$$\lambda_{\text{EC}} = N_g \left(1 + \frac{\tau}{2}\right)^2 + 6 = 14 + 8\tau + 2\tau^2,$$

and the transverse-traceless eigenvalue is

$$\lambda_{\text{TT}} = 14 = 2N_R.$$

The squash amplitude is

$$s_0 = n_{\text{broken}} \tau = 2\tau,$$

with $n_{\text{broken}} = 2$ the number of broken generators of $\mathfrak{su}(2)_R$, and its content ratio is

$$\frac{s_0}{N_R} = \frac{1}{175}.$$

The curvature relation of the Einstein-Cartan geometry is

$$R(\omega) = R_{\text{LC}} - \frac{3}{2} \left(\frac{\tau}{L}\right)^2,$$

with $R_{\text{LC}} = 6/L^2$, so the torsion corrects the curvature at order $\tau^2 \sim 10^{-4}$. These quantities are the geometric inputs to the gauge couplings of Section 5 and the electroweak order parameters of Section 7.

5 Gauge couplings

This section computes the three gauge couplings from the endpoint geometry of Section 3 and the content symmetries of Section 4. The couplings are not fitted to the observed values. They are the closed forms of the geometric reduction of the isometry zero modes, corrected by the content ratios that the companion paper fixes.

5.1 The weak coupling and its conservation law

The $\mathfrak{su}(2)_L$ coupling descends from the Kaluza-Klein reduction of the isometry zero modes on RP^3 . The Killing normalisation gives the bare coupling

$$g_2^2 = 8 \left(\frac{M_G}{M_P}\right)^2 kL^{-3}.$$

The window-edge identity $kLM_G = M_P\sqrt{\pi}$ of Section 3.3 collapses this to

$$g_2^2 = \frac{8\pi}{kL^5}, \quad \alpha_W(M_G) = \frac{g_2^2}{4\pi} = \frac{2}{kL^5},$$

so the weak coupling at the emergence scale is a pure function of the window capacity, with no mass scale entering. This is the bare geometric value. It is corrected by the geometric-dynamics conservation law

$$\frac{1}{\alpha_{\text{SM}}} = \frac{1}{\alpha_W} + \frac{1}{N_c} - \frac{\tau^2\pi}{2}, \quad N_c \left(\frac{1}{\alpha_{\text{SM}}} - \frac{1}{\alpha_W} + \frac{\tau^2\pi}{2} \right) = 1,$$

where $\alpha_W = 2/kL^5$ is the window-capacity weak coupling and the terms $1/N_c$ and $\tau^2\pi/2$ are the colour-number and torsion content ratios. The conservation law is equivalent to the conformal-gauge duality $N_g\xi = 1$ of Section 4.2, and its complete derivation is given in Appendix C.4. The resulting central value is

$$g_2(M_G) = \sqrt{\frac{4\pi}{1/\alpha_W + 1/N_c - \tau^2\pi/2}} = 0.508848,$$

which matches the Standard Model comparison value 0.508845 to +0.0005%. The bare Killing value overshoots by +0.34%, and the two content-ratio terms shift the result to the comparison value without introducing a fitted coefficient.

5.2 The hypercharge coupling

The $U(1)_Y$ coupling descends from the $\mathfrak{su}(2)_R \rightarrow U(1)_R$ breaking by the $J = 2$ squash. The squash rescales the $U(1)$ normalisation by the metric factor

$$\kappa^2(s) = \frac{1+s}{(1-2s)^{5/2}},$$

evaluated at the squash amplitude $s_0 = 2\tau$, which gives $\kappa(2\tau) = 1.13183$. The mixing acts at the breaking scale k_{GUT} , and the coupling then runs down to M_G . A further correction of structural origin enters through the content ratio

$$\delta_{g_1} = -\tau r_{23} \sum Y^2 \Delta_f \xi,$$

where $r_{23} = 3/(10\sqrt{3})$ is the hypercharge-trace hierarchy ratio, and the product $\sum Y^2 \Delta_f \xi = (10/3) \cdot (3/2) \cdot (1/8) = 5/8$ is the hypercharge capacity times the fermion conformal weight times the conformal coupling. This is the same conformal-gauge duality that closes the $1/N_c$ term of Section 5.1. The resulting coupling is

$$g_1(M_G) = 0.60499.$$

Together with g_2 , it determines the fine-structure constant by the two-loop geometric renormalisation group (the same running as $\alpha_s(M_Z)$ of Section 5.3), giving $\alpha^{-1}(M_Z) = 128.208$ against the observed 127.952, a deviation of +0.20%.

5.3 The colour coupling

The colour coupling shares the $\mathfrak{su}(2)$ Killing normalisation at order α^0 , so at the breaking scale the two couplings coincide up to the long-root bifurcation

$$g_3(k_{\text{GUT}}) = g_2(k_{\text{GUT}}) \left(1 + \frac{\alpha_{\text{GUT}}^2}{K}\right),$$

where $\alpha_{\text{GUT}} = g_2(k_{\text{GUT}})^2/(4\pi)$ and

$$K = \frac{J(J+2)}{d} = \frac{8}{3}$$

is the long-root condensation coefficient, the kinetic eigenvalue $J(J+2) = 8$ of the $J = 2$ squash over the internal dimension $d = 3$. The bifurcation is the two-loop self-coupling of $SU(3)$ diluted by the long-root condensation, so g_3 sits above g_2 at the breaking scale. Running both couplings with the two-loop renormalisation group, with the geometric Yukawa $y_0 = 1$ of Section 6.2, gives

$$g_3(M_G) = 0.49776,$$

and the strong coupling at the electroweak scale is

$$\alpha_s(M_Z) = 0.11799,$$

matching the observed 0.1179 to +0.073%.

5.4 The two-loop coefficients

The renormalisation group running of Section 5.3 uses the two-loop gauge beta functions. Their coefficients are not imported from a table. They are computed from the group-theoretic content, the quadratic Casimir and Dynkin index of each representation. The one-loop coefficients are

$$b_1 = \frac{41}{10}, \quad b_2 = -\frac{19}{6}, \quad b_3 = -7,$$

and the two-loop matrix is

$$B = \begin{pmatrix} 199/50 & 27/10 & 44/5 \\ 9/10 & 35/6 & 12 \\ 11/10 & 9/2 & -26 \end{pmatrix}.$$

These reproduce the standard values, but they are derived here from the representation content recorded in Section 4.1, so no external coupling table enters the computation. The top-Yukawa and Higgs-quartic beta functions entering the running are evaluated on the same content (the one-loop $9/2 = 3/2 + N_c$, the gauge terms $-3[C_2(Q_L) + C_2(u_R)]$ per group, and the full two-loop coefficients). Their derivation is given in Appendix C.6. The running is carried out with the geometric Yukawa $y_0 = 1$ as the scalar input, and the scale $k_{\text{GUT}} = M_P \sqrt{\pi}/L_{\text{GUT}}$ with $L_{\text{GUT}} = \sqrt{3}/\tau$ is the isometry-breaking scale of the $J = 2$ squash.

6 Flavour structure

This section computes the fermion masses from the window capacity of Section 3 and the torsion parameter of Section 4. The masses are not a set of free Yukawa couplings. They are the geometric ladder of the extrusion order along the scale flow, fixed by the window and its non-adiabatic corrections. The neutrino sector is a phenomenological extension of this ladder and is recorded in Appendix E.

6.1 The ladder

The generation hierarchy is the extrusion order $n = \{0, 2, 4\}$ of Section 3.5, and the mass ladder is the geometric exponent attached to each rung. The first exponent is the window width reduced by the non-adiabatic correction

$$\alpha_{\text{up}} = kL - 2\tau,$$

the rung spacing is

$$\Delta = 6(1 - n_s)kL_{\text{CMB}},$$

where the tilt decrement is $1 - n_s = \tau w_W$ with $w_W = (1 + 2 + 1 + 3)/(d + 1) = 7/4$ for $d = 3$ (the RP^3 Weyl-window ratio; see App. C.14), and $kL_{\text{CMB}} = kL(1 - \tau/4)$ is the local window at the cosmic-microwave pivot. Below we write the positive decrement as $n_s^{\text{tilt}} \equiv 1 - n_s$. The down and lepton ladders split by the hypercharge ratio

$$\frac{9}{8} = \frac{1}{1 - (Y_d/Y_l)^2},$$

and the rung exponents are

$$\alpha_{\text{dn}} = \alpha_{\text{up}} - \frac{18}{17}\Delta, \quad \alpha_{\text{lp}} = \alpha_{\text{up}} - 2\Delta,$$

with the down first-generation step carrying the colour-dilution correction

$$\alpha_{\text{sd}} = \alpha_{\text{dn}} - \frac{kL_{\text{CMB}}}{6}.$$

These ladder indices are intermediate definitions: each exists only to carry the mass ratios that follow, and their numerical values are fixed in Appendix C.7. The ladder and its colour-dilution step are derived in Appendix C.7.

6.2 The top-quark anchor

The top quark is the anchor of the ladder. Its Yukawa coupling is the $(0, 0)$ diagonal overlap of the $n = 0$ mode, which is exactly normalised

$$y_0 = 1,$$

and it is scale-invariant, in the sense that it does not run along the geometric renormalisation group. The top mass is then fixed by the vacuum expectation value of Section 7,

$$m_t = \frac{y_0 v}{\sqrt{2}} = 174.1 \text{ GeV}.$$

The normalisation $y_0 = 1$ is derived in Appendix C.8.

6.3 The lower masses

The bottom Yukawa is the geometric average

$$\frac{y_b}{y_t} = e^{-(2\alpha_{\text{dn}} - n_s^{\text{tilt}}(kL_{\text{CMB}} + 2\tau))} = 0.024344,$$

where n_s^{tilt} is the positive spectral-tilt decrement. This gives the bottom mass

$$m_b = \frac{y_b}{y_t} m_t = 4.238 \text{ GeV}.$$

The mass ratios within each sector are the geometric exponents

$$\frac{m_s}{m_d} = e^{2\alpha_{\text{sd}}} = 19.62, \quad \frac{m_b}{m_s} = e^{2\alpha_{\text{dn}}} = 44.87, \quad \frac{m_\mu}{m_e} = e^{2\alpha_{\text{lp}} + \sqrt{2}\pi} = 206.36,$$

and the charm and up ratios are

$$\frac{m_t}{m_c} = e^{2\alpha_{\text{up}}} = 135.2, \quad \frac{m_t}{m_u} = e^{2\alpha_{\text{up}}} e^{2kL_{\text{CMB}} + \ln 4} = 77304.$$

The electron mass is the Planck-anchored closed form

$$m_e = M_{\text{P}} e^{-20kL} (1 - s_0\kappa) = 0.510 \text{ MeV},$$

where $s_0\kappa$ is the squash correction of Section 4.4. The down quark, the strange quark, the muon, and the tau follow from these ratios and the sector anchors. The full set of absolute masses descends from the single Planck anchor through the window capacity and the ladder exponents, with no mass inserted by hand. The closed forms of this subsection are derived in Appendix C.9.

7 Electroweak breaking

This section computes the quartic order parameter, the positive colour-singlet scalar mass, and the vacuum expectation value. The CKM and CP content is recorded in Appendix E. These are the quantities that turn the condensation structure of the companion paper into numbers.

7.1 The quartic order parameter

The scalar order parameter is the quartic coupling of the colour-singlet sector. It is fixed by the stationarity of the effective potential at the onset of the symmetry breaking, together with the Einstein-Cartan torsion algebra. The closed form is

$$\lambda = \frac{\xi(R_c - R_{\text{GUT}})}{(2\tau)^2} = 149.145,$$

where $\xi = 1/8$ is the conformal coupling, $R_c = 6/\pi$ is the critical curvature, and $R_{\text{GUT}} = 6/L_{\text{GUT}}^2$ with $L_{\text{GUT}} = \sqrt{3}/\tau$ is the curvature at the breaking onset. The torsion algebra fixes the three coefficients of the parity-preserving torsion Lagrangian as $a = M_{\text{G}}^3/4$, $b = 4a$ by the algebraic-torsion condition, and $c = -(7/3)a$ by the trace term, and the stationarity

$\partial V/\partial\varphi = 0$ then selects the quartic coupling. The value 149 is the dimensionless self-coupling that the stationarity and the torsion algebra produce, with no scalar mass or coupling inserted by hand. This λ is the colour-singlet order parameter. It is distinct from the Higgs quartic λ_H that enters the tree-level Higgs mass $m_H = \sqrt{2\lambda_H} v$ of Section 10.6; λ_H itself is closed by the pseudo-dilaton identity of Appendix C.12. The complete derivation is in Appendix C.11.

7.2 The positive colour-singlet scalar mass

The companion paper derives the positive colour-singlet scalar mass from the condensation of the colour-singlet scalar. The present section records its numerical content. The condensate trigger is the long-root tachyon

$$m_{\text{long}}^2 = K(R - R_c) < 0 \quad \text{for} \quad R < R_c,$$

with $R_c = 6/\pi$. Below the critical curvature the mode condenses, and the lowest colour-singlet excitation acquires the positive squared mass

$$m_\delta^2 = \frac{16}{3}(R_c - R) > 0.$$

At the flow endpoint the condensation condition is realised numerically, not left conditional. The endpoint is $\sigma_C = 6.95 \times 10^{41} \text{ GeV}^{-1}$, the Hubble scale of Section 8.2, and the long-root mode is already tachyonic at the emergence scale, $m_{\text{long}}^2(M_G) = K(R(M_G) - R_c) = -2.52 < 0$. Since $R = 6/L^2$ decreases monotonically along the self-similar flow $L(k) = C/k$, the endpoint curvature satisfies $R(\sigma_C) < R(M_G) < R_c = 6/\pi$, so the endpoint lies in the condensed phase and the positive scalar mass $m_\delta^2 > 0$ is realised there.

The condensate expectation value is fixed by

$$\langle E \rangle^2 = \frac{8}{3} \frac{R_c - R}{\lambda},$$

and the positive scalar mass is $m_\delta^2 = 2\lambda\langle E \rangle^2$. The physical value of this positive scalar mass is the glueball tower of Section 9.2, whose lightest member sits at $8\Lambda_{\text{QCD}}$. The colour-singlet sector is therefore gapped, with no zero mode beyond the constants, and the positive scalar mass is a spectral lower bound rather than a fitted scale.

7.3 The vacuum expectation value

The electroweak vacuum expectation value is the emergence scale times the window-squared line. Its closed form is

$$v = M_G \frac{3\alpha}{\pi} e^{-4\pi k L} (1 - s_0 \kappa) = 246.19 \text{ GeV},$$

where the squash correction $s_0 \kappa$ with $s_0 = 2\tau$ and $\kappa = 1.13183$ is the $J = 2$ isometry-breaking normalisation of Section 5.2. The value matches the observed 246.22 GeV to -0.012% . The hierarchy $v/M_G \sim 10^{-16}$ is the geometric window-squared line, the analogue for the electroweak scale of the gauge-coupling conservation law of Section 5.1, and it is produced by the exponential $e^{-4\pi k L}$ rather than by a fine-tuned scalar mass.

8 Cosmology

This section computes the homogeneous cosmological quantities from the same spectral datum. Two Gaussian entropies, one at the Planck anchor and one at the vacuum floor, fix the entropy integral, and from it the Hubble rate, the dark-energy fraction, and the microwave temperature follow. The endpoint residual is then propagated as a cold source in the linear cosmological comparison. No background cosmological parameter is taken from observation in these closed forms.

8.1 Two Gaussian entropies

The entropy integral of the scale flow is the logarithm of the ratio of two Gaussian widths. The Planck Gaussian has width M_{P}^2 , and the vacuum-floor Gaussian has width $\sqrt{\rho_{\Lambda}}$, where the vacuum energy density is the dark-energy content, a closure relation of the spectral datum

$$\rho_{\Lambda} = Y_u m_{\nu_1}^4 (1 - 4s_0\kappa) = \frac{2}{3} m_{\nu_1}^4 (1 - 4s_0\kappa).$$

The factor $(1 - 4s_0\kappa)$ is the dark-energy-weight symmetry correction of the $J = 2$ squash. The Weinberg mass m_{ν_1} carries $+s_0\kappa$, so its fourth power carries $+4s_0\kappa$, and the weight cancels it to order $s_0^2\kappa^2$, keeping ρ_{Λ} conserved under the squash level transfer (Appendix C.14). The entropy integral is

$$\int \gamma_M = \ln \frac{M_{\text{P}}^2 \sqrt{2\pi + r_{23}}}{\sqrt{\rho_{\Lambda}}} = 139.2537,$$

with $r_{23} = 3/(10\sqrt{3})$ the hypercharge-trace hierarchy ratio. This single number, the logarithm of the window span, is the bridge between the ultraviolet Planck anchor and the infrared vacuum floor, and it encodes the Boltzmann entropy of the flow as $S = \ln W$ with W the window span.

8.2 The Hubble rate and the dark-energy fraction

The Hubble rate is the infrared endpoint of the emergence window,

$$H_0 = M_{\text{P}} \sqrt{\pi} e^{-\int \gamma_M} = 1.4388 \times 10^{-42} \text{ GeV},$$

which corresponds to $67.45 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The dark-energy fraction is

$$\Omega_{\Lambda} = \frac{\rho_{\Lambda}}{3H_0^2 M_{\text{P}}^2} = \frac{2}{3} + \frac{r_{23}}{3\pi} = 0.68504.$$

The second form shows that Ω_{Λ} is a content ratio, the $2/3$ of the maximum-entropy closure plus the hypercharge-trace correction $r_{23}/(3\pi)$. The value matches the Planck determination to $+0.05\%$. The full derivation of this section is in Appendix C.14.

The long-range transverse-traceless sector selected by the zero-momentum spectral pole has the low-energy tensor structure represented by

$$S_{\text{eff}} = \frac{M_{\text{P}}^2}{2} \int d^4x \sqrt{-g} R - \int d^4x \sqrt{-g} \rho_{\Lambda},$$

with $M_{\text{P}}^2 = 1/(8\pi G)$. The entropy integral fixes H_0 , the neutrino floor fixes ρ_Λ , and their ratio gives the dark-energy content fraction

$$\Omega_\Lambda = \frac{\rho_\Lambda}{3H_0^2 M_{\text{P}}^2} = \frac{2}{3} + \frac{r_{23}}{3\pi}.$$

The full endpoint budget gives the corresponding homogeneous Friedmann relation,

$$\rho_{\text{tot}} = \rho_\Lambda + \rho_b + \rho_\Sigma = 3H_0^2 M_{\text{P}}^2, \quad \Omega_b + \Omega_\Sigma + \Omega_\Lambda = 1.$$

Thus the baryon component and endpoint residual complete the standard homogeneous relation. The Friedmann form is the consistency relation among the independently closed values of H_0 , ρ_Λ , ρ_b , and ρ_Σ .

8.3 The microwave background temperature

The cosmic-microwave temperature is the photon floor set by the lightest neutrino. The raw floor is

$$T_{\text{CMB}}^{\text{raw}} = \frac{m_{\nu_1} r_{12}}{\pi} (1 - s_0 \kappa) (1 - \tau \Delta_s) = 2.7310 \text{ K},$$

where $r_{12} = 3/10$ is the hypercharge ratio and $\Delta_s = 1/2$ is the scalar conformal weight, and the factor $(1 - s_0 \kappa)$ is the same squash correction as the electroweak scale. The photon floor carries $-s_0 \kappa$ while the seesaw mass carries $+s_0 \kappa$ (the level transfer). The finite photon zero-mode factor at the endpoint is

$$C_\gamma = 1 - \frac{\tau}{\pi^2}, \quad T_{\text{CMB}} = C_\gamma T_{\text{CMB}}^{\text{raw}} = 2.72547 \text{ K}.$$

The corrected value is used for the CMB-monopole comparison and for the linear comparison propagation. The internal baryon-density closure of Section 8.4 uses the raw photon floor together with the raw Sakharov value of η_B .

8.4 Flatness from the closure relation

The three energy fractions close to unity by the closure relation. The baryon fraction is

$$\Omega_b = \frac{\eta_B^{\text{raw}} n_\gamma^{\text{raw}} m_p}{\rho_{\text{crit}}} = 0.04925,$$

built from the raw Sakharov baryon asymmetry η_B of Appendix E, the raw photon density

$$n_\gamma^{\text{raw}} = \frac{2\zeta(3)}{\pi^2} (T_{\text{CMB}}^{\text{raw}})^3,$$

the proton mass of Section 9.1, and the critical density $\rho_{\text{crit}} = 3H_0^2 M_{\text{P}}^2$ with the Hubble rate of Section 8.2. The internal Ω_b closure uses the raw photon floor and raw Sakharov ratio. The finite endpoint factor C_γ of Section 8.3 supplies the corrected microwave monopole used in the observational comparison. The dark-energy fraction is 0.68504 of Section 8.2, and the remaining fraction is fixed by flatness

$$\Omega_\Sigma = 1 - \Omega_\Lambda - \Omega_b = 0.26570.$$

Table 2. Compressed linear-cosmology quantities obtained by propagating the closed endpoint set with $\Omega_{\text{cdm}} := \Omega_{\Sigma}$ and the corrected photon monopole through CAMB. The propagation is a comparison step only: no CAMB output is used to set a closure parameter. The reionization optical depth used for the CMB spectra is a standard visibility parameter, not a fundamental closure.

Quantity	Comparison value
$\Omega_m = \Omega_b + \Omega_{\Sigma}$	0.314956
r_{drag}	146.980 Mpc
$100 \theta_*$	1.041595
z_{eq}	3414.88
σ_8	0.81423
S_8	0.83428

The sum

$$\Omega_b + \Omega_{\Sigma} + \Omega_{\Lambda} = 1.00000$$

is the flatness closure of the three fractions. The resulting Ω_{Σ} is the maximum-entropy endpoint residual and occupies the cold-matter slot in the linear cosmological propagation. The reference column compares it with the conventional cold-dark-matter fraction. Its local reading, together with the acceleration scale a_0 , is recorded in Appendix G.

8.5 Endpoint residual in linear propagation

The linear endpoint closure propagates the Hamiltonian residual as a conserved cold source,

$$T_{\mu\nu}^{\Sigma} = \rho_{\Sigma} u_{\mu} u_{\nu}, \quad \nabla_{\mu}(\rho_{\Sigma} u^{\mu}) = 0, \quad p_{\Sigma} = 0, \quad c_{s,\Sigma}^2 = 0, \quad \pi_{\Sigma} = 0.$$

In a standard Boltzmann propagation it fills the cold-dark-matter slot as $\Omega_{\text{cdm}} := \Omega_{\Sigma}$. Appendix G records the separate local acceleration projection of the same endpoint. With the closed outputs and corrected photon monopole fixed, CAMB propagates this set to the compressed comparison quantities in Table 2. These quantities are evaluated after the closure chain and enter the comparison layer.

8.6 The gravitational sector

The gravitational sector is the long-range transverse-traceless mode associated with the energy–momentum spectral criterion of Ref. [2]. The gravitational coupling takes the form

$$G_{\text{N}} = \frac{1}{8\pi Z_{\text{phys}} M_{\text{P}}^2},$$

where the matter back-reaction factor is

$$Z_{\text{phys}} = \frac{\lambda_{\text{long}}}{\lambda_{\text{long}} + \sigma} = 1.000000.$$

The transverse-traceless projection fixes the four-dimensional tensor channel, while its compact spectral normalisation is supplied by the long-root eigenvalue of the $J = 2$ squash

mode. This mode is the Spin(4) representation $(j_L, j_R) = (2, 1)$; at the emergence scale,

$$\lambda_{\text{long}} = \frac{C_2(2, 1)}{L^2} = \frac{2[2(2+1) + 1(1+1)]}{L^2} = \frac{16}{L^2} = \frac{16M_G^2}{kL^2}.$$

The quantity σ is the five-channel matter self-energy,

$$\sigma = \left| \sum_{\text{channels}} V_3 \Pi_{\text{channel}}^2 \right| \frac{k^2}{M_P^2},$$

where the five channels are the $T_{\mu\nu}$ spin-2, $T_{\mu\nu}$ spin-0, F^2 , G^2 , and J^μ amplitudes used in the spectral-pole closure of Appendix C.1, each evaluated as a discrete spectral sum over RP^3 . At the emergence scale the ratio is $\sigma/\lambda_{\text{long}} = 1.57 \times 10^{-37}$, so $Z_{\text{phys}} = 1$ to the displayed precision. The self-energy is tiny compared with the eigenvalue, so the back-reaction does not shift the gravitational normalisation. The Newton constant is the single observed anchor of the construction, and $G_N = 1/(8\pi M_P^2)$ is the identity that defines the anchor. With the anchor fixed by the observed value, the residue $Z_{\text{phys}} = 1$ shows that matter back-reaction is negligible at the stated level. The tensor propagation and confinement sectors use their separately defined quantities $p_{\text{TT}}^2 = 8/L^2$ and $\lambda_{\text{TT}} = 14/L^2$.

The low-acceleration scale a_0 that follows from this long-range structure, and the endpoint-residual reading, are recorded in Appendix G.

9 QCD scale and confinement

This section computes the QCD scale, the proton mass, the glueball spectrum, and the confinement quantities. The primordial helium abundance is recorded in Appendix F. The QCD scale descends from the colour coupling of Section 5.3, and the spectral quantities descend from it.

9.1 The QCD scale and the proton mass

The QCD scale is fixed by the two-loop running of the colour coupling $g_3(M_G)$ of Section 5.3 down to the point where the coupling becomes strong, with the top-quark threshold matched at the geometric mass m_t . The extracted scale carries the correction $(1 - s_0\kappa/N_g)$ for the Yukawa difference between the scale-invariant geometric Yukawa $y_0 = 1$ and the running Standard Model top Yukawa in the two-loop gauge beta function, with $s_0 = 2\tau$ the squash amplitude and $N_g = 8$ the generator count. The correction is included explicitly in the scale definition (Appendix C.15). The result is

$$\Lambda_{\text{QCD}} = 0.2074 \text{ GeV},$$

a deviation of -1.25% from the observed 0.210 GeV. The proton mass is the three-quark bound state expressed through the symmetry content, a spectral estimate of the confinement scale

$$m_p = N_c \Delta_f \left(1 - \frac{1}{N_g^2 \Delta_s}\right) \Lambda_{\text{QCD}} \left(1 + \tau\kappa \sum Y^2 \Delta_s\right) = \frac{279}{64} \Lambda_{\text{QCD}} \left(1 + \frac{5\tau\kappa}{3}\right) = 0.9382 \text{ GeV},$$

where $N_c = 3$, $\Delta_f = 3/2$, and the factor $1 - 1/(N_g^2 \Delta_s) = 31/32$ is the conformal-duality correction¹, with the final bracket the scheme correction between the constituent-quark scale and the $\overline{\text{MS}}$ scale. The value matches the observed proton mass to -0.012% .

9.2 The glueball spectrum

The colour-singlet scalar sector of the companion paper has a discrete spectrum whose lightest level is the positive colour-singlet scalar mass. The glueball tower is the geometric ladder

$$\lambda = 2\lambda_{\text{gluon}} + C_2(J) + n N_g \xi,$$

with $N_g \xi = 1$, and the lightest scalar, a spectral estimate of the positive colour-singlet scalar mass, is

$$m_G = \lambda(0^{++})\Lambda_{\text{QCD}} = 8\Lambda_{\text{QCD}} = 1.659 \text{ GeV}.$$

The tower ratios are geometric,

$$2^{++}/0^{++} = \sqrt{2}, \quad 0^{++*}/0^{++} = 1.50, \quad 0^{-+}/0^{++} = 1.46,$$

so the tensor state sits at 2.346 GeV, the first conformal excitation at 2.489 GeV, and the pseudoscalar at 2.418 GeV. The pseudoscalar level 0^{-+} at 2.418 GeV is compared with the quenched lattice-QCD prediction $m_{0^{-+}} = 2560 \text{ MeV}$ [18] and with the glueball-like state $X(2370)$, whose mass $2359_{-14}^{+13} \text{ MeV}$ and spin-parity $J^{PC} = 0^{-+}$ were reported by the BESIII Collaboration in J/ψ radiative decays [19–21]. The level sits 5.5% below the pure-gauge lattice value and 2.5% above the $X(2370)$ mass. The pure-gauge comparison is not a full hadronic calculation: in the physical spectrum the 0^{-+} glueball component can mix with flavour-singlet pseudoscalar channels, including η' and η_c , and such mixing can shift the observed resonance relative to the quenched value. The present value is therefore listed as the pseudoscalar comparison target of the tower, while the scalar 0^{++} level remains a lattice-scale comparison [18]. The $\sqrt{2}$ ratio is the two-gluon tensor over the two-gluon scalar, a pure spectral ratio. This closes the positive colour-singlet scalar mass of Section 7.2 at the numerical level, with this mass given by the lightest glueball rather than by a fitted scale.

9.3 String tension and deconfinement

The confinement quantities are the spectral eigenvalues of the transverse-traceless and vector sectors. The string tension is

$$\sigma = \frac{\lambda_{\text{TT}}}{\pi} \Lambda_{\text{QCD}}^2 = \frac{14}{\pi} \Lambda_{\text{QCD}}^2 = 0.1917 \text{ GeV}^2,$$

where $\lambda_{\text{TT}} = 14$ is the lowest transverse-traceless eigenvalue. The deconfinement temperature is

$$T_d = \frac{\lambda_{\text{vector}}}{N_c} \Lambda_{\text{QCD}} (1 - \tau\kappa) = \frac{4}{3} \Lambda_{\text{QCD}} (1 - \tau\kappa) = 270 \text{ MeV},$$

¹The factor $31/32$ follows from the two identities $N_g \xi = 1$ and $N_g \Delta_s = 2(d-1)$ of Section 4.2, since $1/(N_g^2 \Delta_s) = 1/(8^2 \cdot 1/2) = 1/32 = \xi/4$, with $\Delta_s = (d-2)/2 = 1/2$ the scalar conformal weight. The derivation is given in Appendix C.15.

where $\lambda_{\text{vector}} = 4$ is the lowest vector eigenvalue and $(1 - \tau\kappa)$ the chiral-squash correction, since deconfinement is accompanied by chiral restoration, so the deconfinement scale inherits the chiral-squash content of Section 4.4. The ratio

$$\frac{\sigma}{T_d^2} = \frac{14}{\pi} \cdot \frac{9}{16} (1 - \tau\kappa)^{-2} = \frac{126}{16\pi(1 - \tau\kappa)^2} = 2.6242$$

is the self-consistency of the confinement scale. The content ratio $126/(16\pi) = 2.5068$ between the area law and the deconfinement temperature, softened by the chiral-squash factor $(1 - \tau\kappa)^2$. The full spectral derivation of this subsection is in Appendix C.15.

10 Numerical results and comparison

This section collects the numerical results and compares them with observation. The full parameter table is given in Appendix A, produced from the closed forms of Appendix C rather than stated independently. The present section summarises the table by sector and records the deviation distribution.

The numerical comparisons reported below evaluate a fixed closure chain against reference values introduced in the comparison column. Scheme-sensitive quantities carry their perturbative and renormalisation status. The resulting agreements are correlated consistency checks because many outputs descend from the same intermediates.

The percentage deviations in the tables are signed central-value residuals. A statistical interpretation would additionally require experimental covariance, theory uncertainties, and a specified joint likelihood. The tier labels record the derivational distance and the perturbative or modelling status of each comparison.

10.1 The interface chain

The interface with the Standard Model is formulated as a boundary-condition problem. At the emergence scale M_G the spectral sums fix the ultraviolet boundary conditions: the gauge couplings $g_i(M_G)$ of Section 5, the geometric Yukawa $y_0 = 1$ of Section 6.2, the Higgs quartic λ_H of Appendix C.12, the electroweak scale v of Section 7.3, and the cosmological densities. Below M_G the geometric renormalisation group supplies the scale flow. The gauge couplings run with the two-loop coefficients of Section 5.4, derived from the group-theoretic content, while the geometric Yukawa and Higgs quartic retain the values fixed by the exact overlap and pseudo-dilaton closures. The flow terminates at the internal Z mass M_Z , the self-consistent fixed point of Section 10.6; the resulting m_t , m_b , $\alpha_s(M_Z)$, $\alpha(M_Z)$, and electroweak precision quantities are then compared with observation. Appendix D.2 establishes the identification of the spectral-scale flow with the standard renormalisation-group equations in their shared low-energy regime. Thus the spectral construction supplies the boundary data and the standard equations supply the low-energy transfer map. Figure 3 summarises the chain.

10.2 The parameter table

The computation produces the full set of quantities listed in Appendix A. One of them, the Newton constant G_N , is the observed anchor of Section 2.1. The remainder are computed

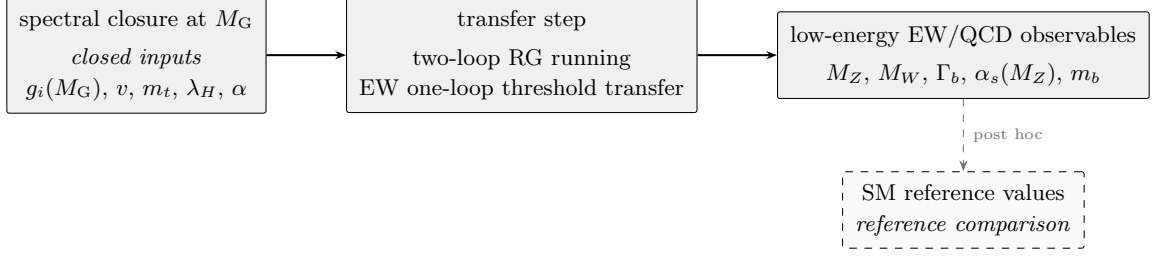


Figure 3. The three-stage interface chain of Section 10.1. The spectral construction supplies the ultraviolet boundary conditions at M_G (left); the transfer step applies the two-loop renormalisation-group running and the electroweak one-loop threshold transfer (centre); the low-energy electroweak and QCD observables emerge for comparison (right). The Standard-Model formulas (including the Denner electroweak reduction) enter only as the transfer map in the centre, never as a source of parameters; the reference values (dashed) enter only at the final comparison.

Table 3. The sectors of the computation and one representative value per sector, with the epistemic status of Table 1. The full table is Appendix A.

Sector	Representative quantity	Computed value	Status
Internal scale	generation count n_g	3	closure
Gauge couplings	weak coupling $g_2(M_G)$	0.50885	derived
Flavour	top mass m_t	174.1 GeV	derived
Electroweak	vacuum expectation v	246.19 GeV	derived
Electroweak precision	Z mass M_Z	91.124 GeV	derived
Cosmology	Hubble rate H_0	67.45 km s ⁻¹ Mpc ⁻¹	exploratory
QCD	proton mass m_p	0.9382 GeV	spectral est.

from the spectral datum, each returned by a closed-form derivation recorded in Appendix C. Only a smaller subset of these quantities are independent comparison channels, however. Many are correlated downstream quantities built from the same few intermediates, and their epistemic status is recorded in Table 1 of Section 2, while the Tier column of Appendix A assigns each comparable quantity to one of the four tiers of Table 4. The words “closure”, “derived”, “scheme-sensitive”, and “exploratory” in the sector summary refer to this status-and-tier classification. The table of Appendix A lists, for each quantity, its closed form, its computed value, the reference value where one exists, and the signed deviation. The sectors and their representative values are collected in Table 3.

10.3 The deviation distribution

The three gauge couplings agree with their reference values to better than 0.01%, the fine-structure constant to 0.20%, the electroweak scale to 0.01%, the charged-fermion mass ratios and neutrino mass-squared splittings to within a percent, the downstream cosmological entries to within a percent, and the QCD scale to about one percent. These signed central-value residuals are evaluated after the closure chain has produced its outputs. Appendix B records numerical stability and the separation of the computation and comparison

Table 4. The four tiers of the comparison. Tier A collects the fixed data of the computation, Tier B the core outputs, Tier C the scheme-sensitive quantities, and Tier D the exploratory downstream extensions. The tiers separate the fixed data from the outputs and the scheme-sensitive quantities from the exploratory extensions, so that distinct comparison channels are not combined into a single undifferentiated list.

Tier	Content	Reading
A	anchor and closure	the fixed data of the computation: the Newton constant, structural constants, and generation count
B	core outputs	gauge couplings, mass ratios, v , electroweak and charged-fermion masses, and the neutrino mass scale and oscillation splittings
C	scheme-sensitive	m_b , Λ_{QCD} , Γ_b , $\sin^2 \theta_{\text{eff}}$, consistency checks labelled by their perturbative and renormalisation status
D	exploratory	PMNS and CP closure relations, the baryon asymmetry, cosmological fractions and T_{CMB} , hadronic scales, and other phenomenological extensions

layers; Figure 4 collects the full distribution.

10.4 Deviations above one percent

Three derived quantities outside the electroweak precision block of Section 10.6 deviate from their observed central values by more than one percent, with the source of each shift identified below. The tensor-to-scalar ratio is compared as an inequality: the value $r = 0.0253$ of Appendix C.14 satisfies the current combined bound $r < 0.036$ [22]. The neutrino comparison uses the oscillation observables Δm_{21}^2 and Δm_{31}^2 directly. The raw light-pair rest split is $6.874 \times 10^{-5} \text{ eV}^2$; the signed table uses the tilt-dressed propagation value of Appendix C.10.

- The bottom mass m_b deviates by +1.38%. It is the geometric average of Section 6.3, and the deviation reflects the scale-invariant anchor $y_0 = 1$ rather than a fitted bottom Yukawa.
- The QCD scale Λ_{QCD} deviates by −1.25%. It is fixed by the two-loop running, and the deviation is the loop-order accuracy of the two-loop against the four-loop extraction.
- The lightest glueball mass deviates by −2.41%. It is the spectral eigenvalue $8\Lambda_{\text{QCD}}$,

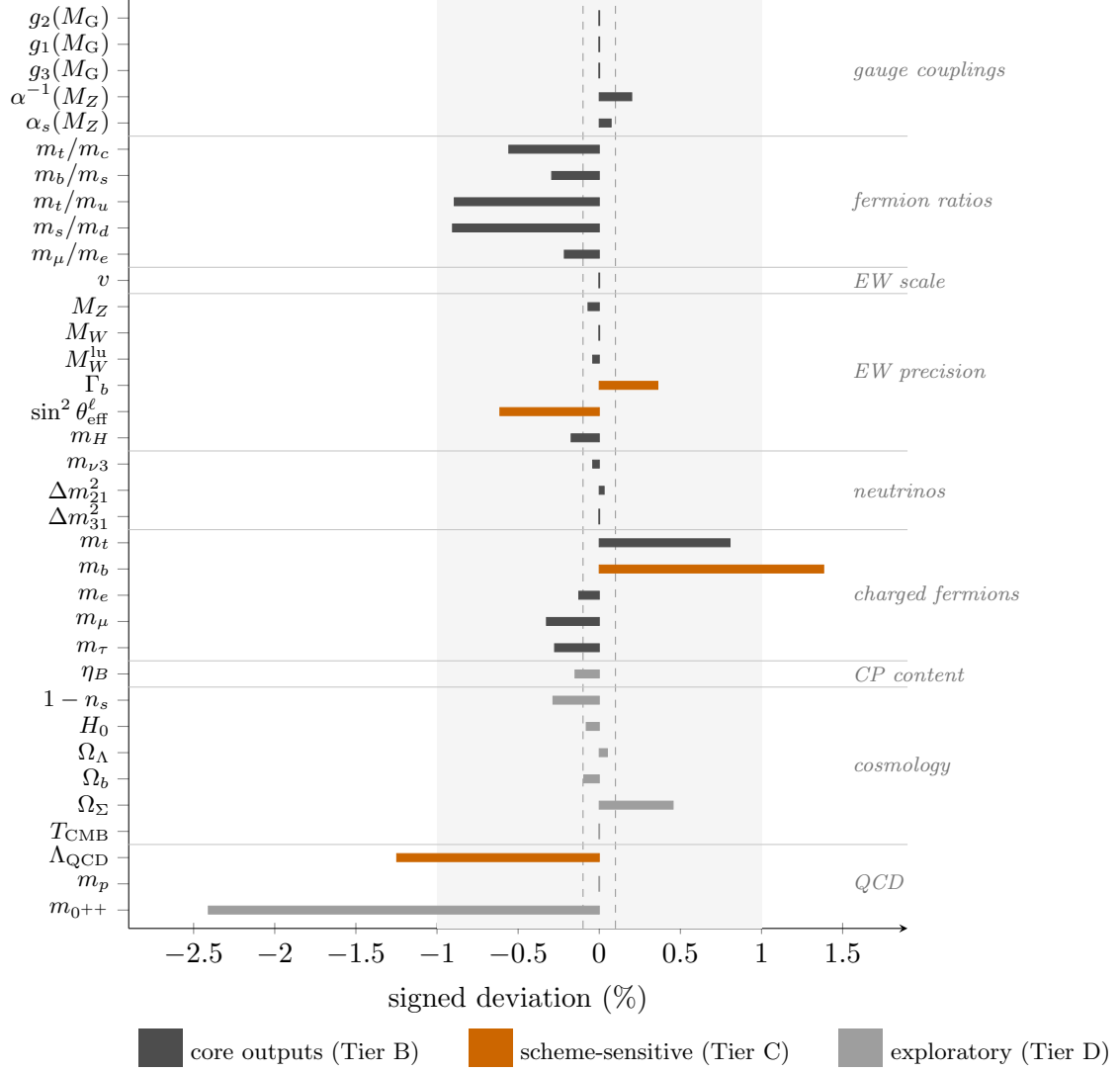


Figure 4. Signed central-value residuals of the comparable quantities of Appendix A, grouped by sector and coloured by the tier of Table 4. The shaded band marks $\pm 1\%$, the inner dashed lines $\pm 0.1\%$. Residuals with $|\delta| < 0.02\%$ are shown as vertical ticks at the origin rather than as zero-width bars. These residuals are not statistical significances. The three entries outside $\pm 1\%$ (m_b , Λ_{QCD} , m_{0++}) are discussed in Section 10.4.

and the deviation combines the loop-order of Λ_{QCD} with the spectral-level approximation of the tower.

The bottom mass and QCD scale are scheme-sensitive computed quantities of Sections 6 and 9; the glueball row is a phenomenological spectral-level comparison. The PMNS mixing angles, CKM phase, and primordial abundances are formal extensions in Appendices E and F. The remaining core quantities agree with their reference values to within one percent at the level of central-value residuals.

The comparatively scheme-sensitive entries, such as m_b , Λ_{QCD} , m_t , M_W , and Γ_b , test

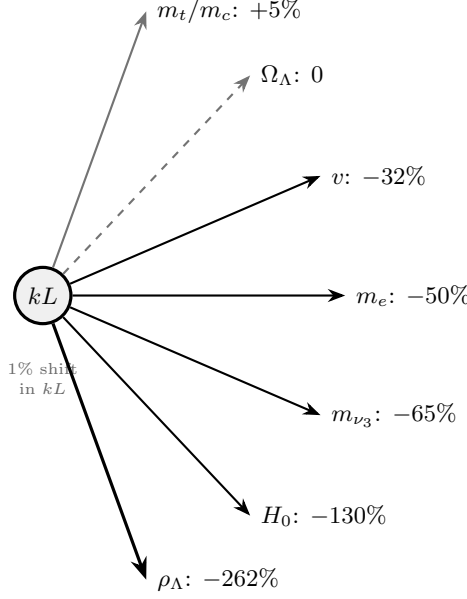


Figure 5. The sensitivity fan of the window capacity kL . A one-percent shift of kL propagates through the exponential hierarchy $e^{-4\pi kL}$, e^{-20kL} , and their powers into the dimensionful outputs, moving v by -32% , m_e by -50% , and ρ_Λ by -262% ; the mass ratios carry only order-unity elasticities, and the dark-energy fraction Ω_Λ , a pure content ratio (dashed, grey), has elasticity exactly zero. The full elasticity matrix is Table 10 of Appendix D.

the same closure data after QCD running, heavy-threshold matching, and electroweak radiative transmission.

10.5 Theoretical sensitivity

Appendix D records the response of the outputs to the structural quantities that carry the chain. The dimensionless hierarchy contains the exponentials $v \propto e^{-4\pi kL}$ and $m_e \propto e^{-20kL}$, so a one-percent shift of kL moves v by -32% , m_e by -50% , and ρ_Λ by -262% . The dark-energy fraction Ω_Λ , a pure content ratio, has zero elasticity. The upstream definitions are the one-loop TT response $K_{\text{TT}} = (1-y)^2$, the endpoint normalisation $\max_{0 \leq y \leq 1} y(1-y)^2 = 4/27$, and the torsion content ratio $\tau = (N_L - N_R)/(N_f \sum Y^2) = 1/50$. Together they fix the closure point, while the elasticity matrix measures the amplification carried by the hierarchy. Figure 5 summarises this response.

10.6 Electroweak precision observables

The interface block of Section 10.1 ends in the electroweak observables measured at high precision. Their computation is a standard electroweak reduction on the closed inputs, namely v , $G_F = 1/(\sqrt{2}v^2)$, $\alpha(M_Z)$ of Section 5, $\alpha(0)$ obtained from the one-loop QED vacuum-polarisation descent of Appendix C.16, the geometric couplings at M_G , the fermion masses of Section 6, and the scale-invariant Higgs quartic λ_H of Appendix C.12. The perturbative level of each quantity is specified below. The numerical values are collected below.

The internal Z mass is the self-consistent fixed point of the tree-level mass formula evaluated on the two-loop geometric running,

$$M_Z = \frac{v}{2} \sqrt{g_2^2 + \frac{3}{5}g_1^2} \Big|_{\mu=M_Z},$$

which gives $M_Z = 91.124$ GeV against the observed 91.1876 GeV (-0.069%), with $\alpha^{-1}(M_Z) = 128.208$ ($+0.20\%$). The W mass is the on-shell relation

$$s^2 c^2 = \frac{\pi\alpha(0)}{\sqrt{2}G_F M_Z^2} \frac{1}{1 - \Delta r}, \quad \Delta r = \Delta\alpha - \frac{c^2}{s^2} \Delta\rho,$$

with $\Delta\alpha = 1 - \alpha(0)/\alpha(M_Z)$ and $\Delta\rho$ the exact one-loop t - b doublet contribution (Veltman). The result is $M_W = 80.371$ GeV against 80.369 GeV ($+0.003\%$), and the on-shell mixing angle $\sin^2 \theta_W = 1 - M_W^2/M_Z^2 = 0.22208$ (-0.50% against the value from the observed masses). The leading universal higher-order correction is the two-loop contribution

$$\Delta\bar{\rho} = \Delta\rho \left[1 + \frac{G_F m_t^2}{8\sqrt{2}\pi^2} (19 - 2\pi^2) \right] = 0.00948$$

(Denner eq. 8.22), together with the resummed relations $\pi\alpha(M_W)/\bar{s}_W^2 = \sqrt{2}G_F M_W^2$ and $\bar{s}_W^2 = s_W^2 + c_W^2 \Delta\bar{\rho}$ (Denner eqs. 8.24 and 8.21). They give $M_W = 80.337$ GeV (-0.040%). Evaluated on conventional Standard-Model benchmark values, the same formulas return $M_W = 80.369$ GeV against the full one-loop reference 80.36 GeV. The level table records the retained leading-universal terms and the residual non-universal contribution Δr_{rem} . The MS-bar-like mixing from the geometric couplings at the internal M_Z is $\sin^2 \theta(M_Z) = 0.23328$ ($+0.89\%$ against the MS-bar value 0.23122). The improved-Born parameter is $\rho = 1/(1 - \Delta\rho) = 1.00951$ against the Standard-Model value 1.00936 ($+0.015\%$).

The Z widths are computed at Born level with the QCD radiator $\alpha_s/\pi + 1.409(\alpha_s/\pi)^2$ for quarks and the QED radiator $3Q_f^2\alpha(M_Z)/(4\pi)$ for charged fermions,

$$\Gamma_f = N_c \frac{G_F M_Z^3}{6\pi\sqrt{2}} (g_{Vf}^2 + g_{Af}^2) (1 + \delta_{\text{QCD}}) (1 + \frac{3Q_f^2}{4\pi} \alpha), \quad g_{Vf} = T_3 - 2Q_f \sin^2 \theta, \quad g_{Af} = T_3.$$

The leading electroweak vertex correction, the top-quark (t, W) loop in $Z \rightarrow b\bar{b}$, is included at its improved-Born value $\delta_b = -G_F m_t^2/(4\sqrt{2}\pi^2) = -(2/3)\Delta\rho = -0.63\%$ (relative width, the standard m_t^2 -enhanced non-universal vertex; cf. Denner section 8.2.3). The calculation retains the leading light-fermion phase-space order; the mass-suppressed terms are $O(m_f^2/M_Z^2) \leq 10^{-3}$. Table 5 states this accuracy level. The total width is $\Gamma_Z = 2.480$ GeV (-0.62%), with the internal M_Z entering through its cube and the retained channel corrections fixing the coefficient. The top-loop vertex reduces Γ_b from 0.3808 to 0.3784 GeV, i.e. from $+1.00\%$ to $+0.36\%$ of the observed value, with the residual offset tied mainly to the closed top mass. The hadronic peak is $\sigma_{\text{had}}^0 = 41.51$ nb ($+0.07\%$). The tree-level Higgs mass $m_H = \sqrt{2\lambda_H} v = 124.98$ GeV (-0.17%), where $\lambda_H = 0.129$ is the Higgs quartic, distinct from the colour-singlet order parameter $\lambda = 149$ of Section 7.1. The absolute muon and tau masses of the internal ladder (105.31 MeV and 1.772 GeV, -0.33% and -0.28%) complete the block.

Table 5. The electroweak precision block of Section 10.1. Every input is a closed value, and the observed values appear only for comparison. The level of each observable is stated in the text. The Z mass is tree-level on the two-loop running, the W mass is at one-loop ρ and at the leading-universal (two-loop ρ) level, Γ_f is at Born plus QCD/QED radiators with the top-loop vertex on Γ_b , $\sin^2 \theta_{\text{eff}}^l$ is at improved Born, and m_H is at tree level.

Quantity	Computed	Observed	Deviation
M_Z (GeV)	91.124	91.1876	−0.069%
$\alpha^{-1}(M_Z)$	128.208	127.952	+0.20%
$\sin^2 \theta(M_Z)$	0.23328	0.23122	+0.89%
M_W (GeV)	80.371	80.369	+0.003%
M_W leading-univ. (GeV)	80.337	80.369	−0.040%
$\sin^2 \theta_W$ (on-shell)	0.22208	0.22321	−0.50%
ρ (improved Born)	1.00951	1.00936	+0.015%
$\sin^2 \theta_{\text{eff}}^l$	0.23011	0.23153	−0.61%
Γ_Z (GeV)	2.4798	2.4952	−0.62%
Γ_{had} (GeV)	1.7331	1.7444	−0.65%
Γ_b (GeV)	0.3784	0.3771	+0.36%
Γ_l (GeV)	0.2499	0.2519	−0.78%
Γ_{inv} (GeV)	0.4967	0.4990	−0.45%
σ_{had}^0 (nb)	41.51	41.481	+0.07%
R_l	20.80	20.767	+0.17%
R_b	0.2197	0.21629	+1.59%
m_H (GeV)	124.98	125.20	−0.17%

Table 6. The perturbative level of each electroweak precision observable in Table 5, together with the leading correction at the next level.

Observable	Evaluated level	Leading next correction
M_Z	tree level on the two-loop running	one-loop electroweak self-energy
M_W	one-loop ρ + leading universal (two-loop ρ , resummed α)	non-universal Δr_{rem} (bosonic, box), ≈ 0.03 GeV
$\sin^2 \theta(M_Z)$	geometric couplings at M_Z	full one-loop $\sin^2 \theta_{\text{eff}}$
$\sin^2 \theta_{\text{eff}}^l$	improved Born (two-loop ρ)	process-dependent κ_l , higher order
Γ_f	Born + QCD/QED radiator, top vertex on Γ_b	light-fermion phase space, other-channel EW vertices
m_H	tree level	one-loop

11 Discussion and conclusion

This paper has evaluated the spectral construction whose structural content is recorded in the companion paper [1]. The calculation begins with the Laplace and Dirac spectrum of RP^3 , the structural constants and closure prescriptions of Section 2, and the Newton constant as its dimensional anchor. Appendix B records the resulting output set, while Appendix D quantifies its dependence on the discrete and functional prescriptions.

The most direct outputs are the window-capacity fixed point $kL = 2.49353$, the associated generation count, the torsion modulus $\tau = 1/50$, and the effective gauge-coupling normalisations. The fermion mass ratios, electroweak scale, and low-energy scales descend from these quantities in an acyclic chain, with G_N converting the dimensionless output into physical units. Their comparison with reference values therefore tests the specified window, pole, and matching prescriptions as a single correlated construction.

The Standard-Model field content is recovered from the spectral datum through zero-mode projection. The gauge fields are the Kaluza–Klein zero modes of the internal connection, their kinetic terms are the zero-mode projection of the field-strength square, and their self-interaction vertices are the corresponding cubic and quartic overlaps. The fermions are the retained spinor modes, with kinetic terms supplied by the restricted Dirac operator and gauge interactions by the connection–current overlap. The scalar is the trace-zero mode, with its kinetic term supplied by the projected Laplacian and its Yukawa couplings by two-spinor–scalar overlap integrals. This is the same overlap structure O_e that enters the charged-fermion masses of Section C.9. The resulting four-dimensional Lagrangian has the Standard-Model form: the zero modes supply the fields and the overlap integrals supply the couplings.

The comparison reports signed central-value residuals. The three gauge couplings agree with their reference values to better than 0.01%, the fine-structure constant to 0.20%, the electroweak scale to 0.01%, and the electroweak precision observables of Section 10.6 to within a percent. The fermion mass ratios, neutrino mass-squared splittings, and downstream cosmological entries also lie within a percent, while the QCD scale differs by about one percent. The larger residuals trace respectively to the scale-invariant bottom anchor, the two-loop QCD running, and the spectral-level approximation of the lightest glueball. The neutrino comparison uses the absolute mass texture together with the finite spectral-tilt propagation relation of Appendix C.10.

The core computation closes at the quantities of Sections 5, 6, 7, and 9. The absolute neutrino scale and its two oscillation splittings are Tier B outputs of the same numerical chain. The homogeneous cosmological quantities of Section 8 are downstream closure evaluations, and their compressed propagation entries are comparison diagnostics. The PMNS and CP relations and the primordial abundances are the formal Tier D extensions recorded in Appendices E and F.

The dimensionful hierarchy is carried by exponentials of the window capacity and is correspondingly sensitive to kL . A one-percent counterfactual displacement of the endpoint closure moves the electroweak scale by -32% , the electron mass by -50% , and the dark-energy density by -262% , while the content ratio Ω_Λ has zero elasticity. Appendix D

Table 7. Near-term comparison targets. The first four rows are direct tests of closed quantities. The PMNS phase, glueball, and acceleration-scale rows test downstream or phenomenological closures, and therefore carry the lower status stated in the text.

Quantity	Value fixed here	Discriminating measurement
Tensor-to-scalar ratio	$r = 0.0253$	a robust upper bound well below this value
Neutrino mass ordering	normal	inverted ordering
Atmospheric octant	$\sin^2 \theta_{23} = 0.551$	lower octant
Summed neutrino mass	$\sum m_\nu = 0.0615 \text{ eV}$	a robust upper bound below this value
PMNS phase (formal closure)	$\delta_{\text{PMNS}} \simeq (8/7)\pi$	value far from this branch
Pseudoscalar glueball target	$m_{0^-+} = 2.418 \text{ GeV}$	no compatible flavour-singlet 0^{-+} state
Low-acceleration branch	$a_0 = 1.204 \times 10^{-10} \text{ m s}^{-2}$, $\mu(y) = y/\sqrt{1+y^2}$	rotation-curve or lensing tests incompatible with the fixed response

traces these responses to the TT kernel, the pole normalisation, and the Gaussian window prescription. The resulting amplification quantifies the rigidity of the adopted closure.

Beyond the individual numbers, the computation produces structural relations among quantities left independent in the Standard Model. The baryon fraction, dark-energy fraction, and endpoint residual close to unity, with the residual occupying the cold-source slot in the linear cosmological comparison. The gravitational coupling is carried by the long-range transverse-traceless mode selected by the zero-momentum spectral pole, with unit back-reaction factor. The glueball tower of Section 9.2 gives the additional target $m_{0^-+} = 2.418 \text{ GeV}$, compared with the $X(2370)$ mass of 2359 MeV reported by BESIII [19, 20]. Each relation carries the dynamical status assigned in its defining section.

Several closed quantities are accessible to current or planned experiments. Table 7 lists their fixed values and the corresponding discriminating measurements. The tensor-to-scalar ratio, neutrino ordering, atmospheric octant, and summed neutrino mass give the most direct tests. The PMNS phase belongs to the formal flavour closure, while the glueball and acceleration-scale entries are phenomenological comparisons with the status stated in Sections 9.2 and G.2.

Appendix G derives the acceleration residue $a_0 = cH_0\sqrt{4/3}/(2\pi)$ and its fixed local response $\mu(y) = y/\sqrt{1+y^2}$ from the maximum-entropy endpoint normalisation. The SPARC and cluster entries compare the fixed quasi-static branch with present data, while the compressed CMB and BAO entries test the conserved linear cold-source branch.

The companion paper defines the admissible structural closure class and supplies the

internal geometry, gauge algebra, fermion content, and positive colour-singlet scalar scale. The present paper evaluates the resulting numerical chain. Its most direct outputs are the window-capacity fixed point, generation count, and gauge-coupling normalisations; the flavour, QCD, and cosmological applications add the effective relations stated in their respective sections. This organization separates the structural classification from its numerical and phenomenological consequences.

A The parameter table

The computation is carried by a small set of structural constants and a set of quantities that admit comparison with observation. The structural constants are the closure fixed point and the integer content identities of the construction. They carry no reference value of their own, but every comparable quantity descends from them, and they are recorded in Table 8. The comparable quantities are listed in Table 9, produced from the closed forms of Appendix C rather than stated independently. Each row gives the quantity, its computed value, the reference value where one exists, the signed deviation, a short record of the derivation, and the tier (A–D) defined in Table 4 of Section 10. One quantity, the Newton constant, is the observed anchor of Section 2.1. The remaining quantities are computed, and none is fitted. The computed values are conditional on the structural choices recorded in the companion paper and on the closure prescriptions of Section 2. The comparison reports the signed central-value deviations of these values against reference values that are kept outside the closure chain and are never inputs to the computation. These signed percentages are bookkeeping residuals for comparison only; they do not include experimental covariance matrices, theoretical scheme uncertainties, or a look-elsewhere correction, and should not be read as statistical significances. Where a direct measurement exists the reference value is the Particle Data Group central value. Where the comparison is made at the emergence scale, as for the gauge couplings $g_i(M_G)$, the reference value is the two-loop Standard-Model extrapolation used in this paper, so the two sides are compared at the same scale.

The derived quantities that are not tabulated here fall into three groups. The intermediate quantities of the chain, namely the ladder indices, the sector steps, the dilaton line, and the correction factors, are auxiliary definitions. Each exists only to carry a later quantity, and each is computed in the derivation of Appendix C where it first appears rather than given a row of its own. The downstream quantities, namely the full electroweak precision block beyond its headline, the cosmological constant, the curvature amplitude, the tensor ratio, the acceleration scale, the string tension, and the deconfinement temperature, are computed in their derivations in Appendix C and in Section 10.6, but are not given rows here. The flavour- and cosmological-extension relations that are stated as structural closure rather than as precision comparison, namely the PMNS mixing angles, the CKM phase and the Jarlskog invariant, and the primordial abundances, are recorded as formal closure in Appendices E and F, where their closure forms are given without a decimal evaluation. The supplementary material lists the additional computed intermediates that are not displayed in this table.

Table 8. The structural constants of the computation. They carry no reference value of their own. Every comparable quantity of Table 9 descends from them together with the single observed anchor.

Structural constant	Value	Origin
window capacity kL	2.49353	closure fixed point, Sec. 3.4
emergence scale M_G	1.731×10^{18} GeV	$M_G = M_P \sqrt{\pi}/kL$, Sec. 3.3
torsion modulus τ	1/50	content ratio, Sec. 4.3
left-handed count N_L	8	$N_L = N_g = N_c^2 - 1$, Sec. 4.1
right-handed count N_R	7	$N_R = N_g - 1$, Sec. 4.1
chiral asymmetry $N_L - N_R$	1	$\pi_1(RP^3) = \mathbb{Z}_2$, Sec. 4.1
hypercharge moment $\sum Y^2$	10/3	anomaly cancellation, Sec. 4.1

Table 9: The comparable quantities of the computation, grouped by sector. Computed values are conditional on the closure prescriptions of Section 2; reference values enter the post-computation comparison layer. Appendix C records the derivation of every closed form.

Quantity	Computed	Observed	Deviation	Tier	Derivation
<i>Internal scale</i>					
generation count n_g	3	3	+0.000%	A	spectral-capacity map on the retained Dirac positions (App. C.1)
<i>Gauge couplings</i>					
weak coupling $g_2(M_G)$	0.508848	0.508845	+0.001%	B	geometric weak coupling after colour and torsion screening (App. C.4)
hypercharge coupling $g_1(M_G)$	0.60499	0.605001	-0.002%	B	squashed-isometry normalisation transported from its scale (App. C.4)
colour coupling $g_3(M_G)$	0.49776	0.497765	-0.001%	B	common-origin colour coupling with conformal correction (App. C.4)
$\alpha^{-1}(M_Z)$	128.209	127.952	+0.200%	B	two-loop running from the emergence boundary (App. C.4)
$\alpha_s(M_Z)$	0.117986	0.1179	+0.073%	B	two-loop gauge flow with derived content coefficients (App. C.5)

Table 9: The comparable quantities of the computation, grouped by sector. Computed values are conditional on the closure prescriptions of Section 2; reference values enter the post-computation comparison layer. Appendix C records the derivation of every closed form.

Quantity	Computed	Observed	Deviation	Tier	Derivation
<i>Fermion mass ratios</i>					
top to charm m_t/m_c	135.242	136	-0.557%	B	up-sector Landau–Zener ladder ratio (App. C.7)
bottom to strange m_b/m_s	44.8679	45	-0.293%	B	down-sector Landau–Zener ladder ratio (App. C.7)
top to up m_t/m_u	7.730364e+04	7.800000e+04	-0.893%	B	up-sector ladder with first-generation window transfer (App. C.7)
strange to down m_s/m_d	19.623	19.802	-0.904%	B	down-sector first-generation ladder ratio (App. C.7)
muon to electron m_μ/m_e	206.356	206.8	-0.215%	B	charged-lepton ladder with the Euclidean-period shift (App. C.9)
<i>Electroweak scale</i>					
Higgs vacuum expectation v (GeV)	246.19	246.22	-0.012%	B	Gaussian window hierarchy at the spectral fixed point (App. C.11)
<i>Electroweak precision</i>					
Z mass M_Z (GeV)	91.1243	91.1876	-0.069%	B	self-consistent neutral electroweak mass relation (Sec. 10.6)
W mass M_W (GeV)	80.3712	80.369	+0.003%	B	on-shell relation with the one-loop rho correction (Sec. 10.6)
W mass, leading-univ. M_W^{lu} (GeV)	80.3369	80.369	-0.040%	B	leading-universal on-shell electroweak relation (Sec. 10.6)

Table 9: The comparable quantities of the computation, grouped by sector. Computed values are conditional on the closure prescriptions of Section 2; reference values enter the post-computation comparison layer. Appendix C records the derivation of every closed form.

Quantity	Computed	Observed	Deviation	Tier	Derivation
Γ_b (GeV)	0.37841	0.37705	+0.361%	C	radiator-improved width with the top vertex term (Sec. 10.6)
$\sin^2 \theta_{\text{eff}}^l$	0.230114	0.23153	-0.612%	C	improved-Born leptonic effective angle (Sec. 10.6)
Higgs mass m_H (GeV)	124.983	125.2	-0.173%	B	tree-level Higgs mass from the closed quartic and scale (Sec. 10.6)
<i>Neutrinos</i>					
heaviest neutrino scale $m_{\nu 3}$ (eV)	0.0501797	0.0502	-0.040%	B	Weinberg two-period texture; atmospheric reference scale (App. C.10)
solar split- ting Δm_{21}^2 (eV ²)	7.412190e-05	7.410000e-05	+0.030%	B	tilt-dressed solar split from the light mass pair (App. C.10)
atmospheric splitting Δm_{31}^2 (eV ²)	0.00251121	0.002511	+0.008%	B	atmospheric split from the absolute mass eigenvalues (App. C.10)
lightest neutrino $m_{\nu 1}$ (eV)	0.00260742	–	–	B	absolute vacuum-floor mass from the hypercharge texture (App. C.10)
<i>Charged-fermion masses</i>					
top mass m_t (GeV)	174.082	172.69	+0.806%	B	geometric unit overlap at the electroweak scale (App. C.8)
bottom mass m_b (GeV)	4.23779	4.18	+1.383%	C	absolute bottom mass from the closed down-sector ratio (App. C.9)
electron mass m_e (MeV)	0.510354	0.511	-0.126%	B	four-level electron cascade over one-generation content (App. C.9)
muon mass m_μ (GeV)	0.105314	0.105658	-0.325%	B	electron mass propagated through the lepton ladder (App. C.9)

Table 9: The comparable quantities of the computation, grouped by sector. Computed values are conditional on the closure prescriptions of Section 2; reference values enter the post-computation comparison layer. Appendix C records the derivation of every closed form.

Quantity	Computed	Observed	Deviation	Tier	Derivation
tau mass m_τ (GeV)	1.77211	1.777	-0.275%	B	muon mass propagated through the lepton ladder (App. C.9)
<i>CP content</i>					
baryon asymmetry η_B	6.090893e-10	6.100000e-10	-0.149%	D	Jarlskog and weak-content Sakharov relation (App. C.13)
<i>Cosmology</i>					
scalar tilt $1 - n_s$	0.035	0.0351	-0.285%	D	torsion modulus times the seven-over-four content ratio (App. C.14)
Hubble rate H_0 (GeV)	1.438850e-42	1.440000e-42	-0.080%	D	infrared endpoint of the maximum-entropy integral (App. C.14)
dark-energy fraction Ω_Λ	0.685044	0.6847	+0.050%	D	dark-energy density divided by the critical density (App. C.14)
baryon fraction Ω_b	0.0492531	0.0493	-0.095%	D	raw photon floor, Sakharov ratio, and proton density (App. C.14)
matter remainder Ω_Σ	0.265703	0.2645	+0.455%	D	maximum-entropy endpoint residual used as a cold source (App. C.14)
CMB temperature T_{CMB} (K)	2.72547	2.72548	-0.000%	D	endpoint-corrected microwave monopole (App. C.14)
<i>QCD</i>					
QCD scale Λ_{QCD} (GeV)	0.20738	0.21	-1.248%	C	two-loop QCD running with top-threshold matching (App. C.15)

Table 9: The comparable quantities of the computation, grouped by sector. Computed values are conditional on the closure prescriptions of Section 2; reference values enter the post-computation comparison layer. Appendix C records the derivation of every closed form.

Quantity	Computed	Observed	Deviation	Tier	Derivation
proton mass m_p (GeV)	0.938156	0.938272	-0.012%	D	conformal proton relation with constituent dressing (App. C.15)
lightest glueball m_G (GeV)	1.65904	1.7	-2.409%	D	lowest colour-singlet spectral level over the QCD scale (App. C.15)

B Reproducibility

Every quantity in the closure chain is returned by a closed-form derivation recorded in Appendix C. The dependencies form an acyclic order from the structural numbers and the Newton anchor to the computed outputs.

The numerical chain contains two input classes: the Newton anchor of Section 2.1 and quantities derived from earlier entries by relations recorded in this paper. Observational reference values and standard propagation tools enter subsequently in the comparison layer.

The parameter table of Appendix A is evaluated from these closed forms. The numerical constants are 2, π , $3/2$, and $1/8$, together with the geometric factors $\sqrt{2\pi}$ and $\sqrt{3/\tau}$, and the observed anchor G_N . The quoted CAMB, BAO, and SPARC entries propagate the fixed outputs into the comparison layer.

The supplementary materials give the algebraic chain in dependency order, together with the numerical values behind the tables. The prediction chain uses the internally computed quantities; observational reference values enter the subsequent comparison layer. The weak-coupling comparison of Section C.1 inverts the conservation law of Section C.4 at the Standard Model reference value of $g_2(M_G)$ after the spectral value of kL has been fixed, and returns the deviation $+0.0002\%$ recorded in the text.

Appendix C gives the algebraic derivations of the identities that carry the quantitative content of the paper, written in the standard mathematical language of the text.

The numerical precision of the window capacity is important because $kL = 2.493534\dots$ is the central intermediate of the chain. It enters both the finite spectral capacity through $(kL)^2 = 6.2177$ of Section 3.5 and the exponential hierarchy through factors of the form $e^{-a kL}$, with $a = 4\pi$ for the electroweak scale and $a = 20$ for the electron mass. A small absolute perturbation δkL therefore induces the relative shift $\delta X/X \simeq -a \delta kL$ in any observable controlled by $e^{-a kL}$. With $a = 20$, rounding kL at the third decimal moves the downstream value at the percent level. The numerical chain therefore propagates the full value, while $kL = 2.49353$ is used for display. Its proximity to the Gaussian scale $\sqrt{2\pi}$ is an

a posteriori numerical relation. The same closed window parameter fixes the finite spectral capacity and the exponential hierarchy scale, and its full precision controls the stability of the reconstruction.

Three reproducibility requirements are imposed. First, re-evaluating the chain returns the core parameters (kL , M_G , v , M_W , Γ_b , and Λ_{QCD}) unchanged within the displayed precision. Second, the observational reference values are held outside the closure chain and are used only after the computation has produced its outputs. Third, central closure values are propagated at full numerical precision, with rounding only at the display stage. Propagations using CAMB/DESI/SPARC enter only as comparisons with the closure parameters held fixed.

B.1 Scope of the formal verification

The optional Lean 4 formalisation supplied as supplementary material covers the discrete algebraic identities used in the construction before the numerical comparison stage. It covers integer counting relations, representation-content identities, normalisation factors, and closure equalities that enter the structural datum of the computation.

The formalisation verifies the stated implications from their declared premises: integer counts, representation-content identities, normalisation factors, and closure equalities. Appendix C supplies the physical derivations, and Section 10 supplies the empirical comparisons.

C Key derivations

This appendix records the derivations that carry the quantitative content of the paper. Each subsection gives the algebra for one closure, so that every closed form in Sections 3–9 is specified within the present paper. The derivations use only the discrete spectrum of RP^3 , the structural constants listed in Section 2, and the measured Newton constant as the single dimensional anchor.

The subsections follow the order of the main text and of the extensions, and each is the derivation behind the section or appendix of the same name. C.1–C.2 close the window capacity and the torsion modulus (Section 3, 4). C.3 records the geometric dynamics. C.4–C.6 derive the gauge couplings and their beta functions (Section 5). C.7–C.9 derive the flavour ladder, the top anchor, and the fermion masses (Section 6), and C.10 the neutrino masses (Appendix E). C.11 derives the electroweak order parameters (Section 7) and C.13 the CKM and CP content (Appendix E). C.14 derives the cosmological quantities (Section 8). C.15 derives the QCD scales (Section 9), and C.16 the primordial abundances (Appendix F).

C.1 The window-capacity closure

The window capacity kL is the fixed point of the spectral-pole condition of the spin-2 channel introduced in Ref. [2]. The object is the transverse-traceless projection of the improved energy-momentum tensor, evaluated as a spectral sum over the discrete RP^3

spectrum. The amplitude at zero external momentum is

$$\Pi_{\mu\nu}^2 = \frac{1}{V_3} \sum_{\text{fields}} \sum_{\text{modes}} d_n w(\text{field}) \int_0^\infty \frac{d\omega}{\pi} K_{\text{TT}}(\omega^2 + \lambda_n, m_{\text{eff}}^2) e^{-(\omega^2 + \lambda_n)/\Lambda^2}, \quad (\text{C.1})$$

where $V_3 = \pi^2 L^3$ is the internal volume, d_n the mode multiplicity, $w(\text{field})$ the channel weight, and Λ the running cutoff in Planck units. The transverse-traceless kernel is

$$K_{\text{TT}}(k^2, m^2) = \frac{k^4}{(k^2 + m^2)^2}. \quad (\text{C.2})$$

The kernel satisfies $K_{\text{TT}}(k^2, 0) = 1$ and is positive for $k^2, m^2 \geq 0$. The Ward-normalised flat-space reference is the associated subtracted amplitude. The channel weights are $w = 4$ for each gauge polarisation, $w = 4$ for each Weyl fermion, and $w = 1$ for each real scalar. The effective masses m_{eff}^2 are the curvature and torsion shifts of each species. The scalar carries the conformal mass $m^2 = \xi R = 3/(4L^2)$ with $\xi = (d-2)/(4(d-1)) = 1/8$ and $R = 6/L^2$. The gauge fields carry $m^2 = C_2/(2L^2) + \tau^2/(6L^2)$, with $C_2 = 3, 2, 0$ for the $\text{SU}(3), \text{SU}(2), \text{U}(1)$ generators. The Weyl fermions carry the torsion shift $m^2 = 3\tau^2/(8L^2)$, while the curvature contribution is included in the Dirac-squared spectrum through $\not{D}^2 = \nabla^* \nabla + R/4$. The transverse-traceless tensor carries the Lichnerowicz shift $m^2 = 6/L^2$. Since $L = k L M_G / k$ along the self-similar flow, the pole condition is an implicit equation in kL . The exponential factor is the Gaussian window of the coarse-graining envelope.

The numerical chain uses the Gaussian heat-flow envelope specified by the maximum-entropy and unbiased coarse-graining conditions, together with semigroup composition, reflection positivity, and scale covariance. Sharp and Litim-type profiles provide counterfactual diagnostics of the dependence on this closure prescription.

The spectral-pole condition selects the emergence scale at which the long-range transverse-traceless response develops its zero-momentum pole. The fixed-point equation is

$$\frac{V_3 \Pi_{\mu\nu}^2(kL, k^2/M_P^2)}{32\pi^2} = \frac{4}{27} \quad \text{at} \quad k = M_G. \quad (\text{C.3})$$

The threshold $4/27$ is the normalisation extremum of the massless-pole condition,

$$\frac{4}{27} = \max_{0 \leq y \leq 1} y(1-y)^2 \quad \text{at} \quad y = \frac{1}{3}, \quad (\text{C.4})$$

where $y = m^2/(k^2 + m^2)$ is the mass fraction of the transverse-traceless kernel. The kernel is $K_{\text{TT}} = k^4/(k^2 + m^2)^2 = (1-y)^2$, and the mass-weighted spectral density is $yK_{\text{TT}} = y(1-y)^2$. Its derivative is $(1-y)(1-3y)$, so the unique interior maximum occurs at $y = 1/3$ and has value $4/27$. The endpoint closure adopts this extremum as the spin-2 pole normalisation. On the self-similar trajectory $L(k) = C/k$ with $C = M_P \sqrt{\pi}$, the volume and cutoff depend on the single combination kL , and the pole condition is one equation in one unknown. Bisection on $[0.3 M_G, 3 M_G]$, followed by the self-consistency iteration between $M_G = C/kL$ and the pole scale k_* , gives

$$kL = 2.49353. \quad (\text{C.5})$$

The closure is cross-checked against the coupling conservation law of Section C.4. After the spectral-pole condition has fixed kL , inversion of the window-capacity form $\alpha_W = 2/kL^5$ at the reference weak coupling $g_2(M_G) = 0.508845$ returns the same value to within 0.0002%. The endpoint geometry $L_{C_g} = \sqrt{\pi}$ fixes the critical curvature $R_c = 6/\pi$, and the window capacity $(kL)^2 = 6.2177$ determines the generation count of Section 3.5.

C.2 The torsion window cancellation

The torsion modulus is the normalised chiral-content invariant of the retained fermion sector. The left-handed content is $N_L = 6 + 2 = 8$, the right-handed content is $N_R = 3 + 3 + 1 = 7$, and the total number of Weyl components is $N_f = 15$. The anomaly-free hypercharge assignment has the quadratic moment

$$\sum Y^2 = 6\left(\frac{1}{6}\right)^2 + 3\left(\frac{2}{3}\right)^2 + 3\left(\frac{1}{3}\right)^2 + 2\left(\frac{1}{2}\right)^2 + 1 = \frac{10}{3}. \quad (\text{C.6})$$

The closure normalises the net chiral excess by the total fermion content and this first non-zero hypercharge moment:

$$\tau = \frac{N_L - N_R}{N_f \sum Y^2} = \frac{8 - 7}{15(10/3)} = \frac{1}{50}. \quad (\text{C.7})$$

This dimensionless content ratio is the torsion modulus used in the spectral shifts and in the squash response below. The $\mathbb{Z}_2$ spin structure supplies the parity class of the chiral excess, while anomaly cancellation supplies the hypercharge normalisation.

C.3 Geometric dynamics

The torsion enters the physical content through the geometric quantities of the transverse-traceless sector. The $J = 2$ Lichnerowicz eigenvalue of the Einstein-Cartan sector is

$$\lambda_{\text{EC}} = N_g \left(1 + \frac{\tau}{2}\right)^2 + 6 = 14 + 8\tau + 2\tau^2, \quad (\text{C.8})$$

the $\mathfrak{su}(2)_L$ spin connection $8(1 + \tau/2)^2$ dressed by the torsion plus the Lichnerowicz shift 6, and the transverse-traceless eigenvalue is

$$\lambda_{\text{TT}} = 14 = 2N_R, \quad (\text{C.9})$$

the content double of the seven right-handed singlets. The squash amplitude is the broken-generator content,

$$s_0 = n_{\text{broken}} \tau = 2\tau, \quad (\text{C.10})$$

with $n_{\text{broken}} = \dim \text{SU}(2)_R - \dim \text{U}(1)_R = 3 - 1 = 2$ the two broken generators absorbed as the Goldstone directions of the W_R bosons, each contributing the torsion modulus. Its content ratio is

$$\frac{s_0}{N_R} = \frac{n_{\text{broken}}}{N_f \sum Y^2 N_R} = \frac{2}{15 \cdot (10/3) \cdot 7} = \frac{1}{175}, \quad (\text{C.11})$$

half the $J = 2$ first-order torsion shift $N_g \tau / 14 = 8\tau / 14$. The 2π Euclidean period appears throughout the construction as a common factor of the ultraviolet and infrared closures, in the amplitude $e^{1/(2\pi)}$, the window width $2L = \sqrt{2\pi}$, and the tensor ratio $r = (1/2\pi)^2$.

C.4 The gauge-coupling conservation law

The weak coupling at the emergence scale is the Killing normalisation of the isometry zero modes. The raw coupling is $g_2^2 = 16\pi^2/I_{kv}$ with the Killing integral $I_{kv} = 2\pi^2 L^3$, and the four-dimensional reduction inserts the normalisation factor $(M_G/M_P)^2$. At the emergence scale $L = kL$ this gives

$$g_2^2 = \frac{16\pi^2(M_G/M_P)^2}{2\pi^2 k L^3} = 8 \left(\frac{M_G}{M_P} \right)^2 k L^{-3}. \quad (\text{C.12})$$

The window-edge identity $kLM_G = M_P\sqrt{\pi}$ collapses this to

$$g_2^2 = \frac{8\pi}{kL^5}, \quad \alpha_W = \frac{g_2^2}{4\pi} = \frac{2}{kL^5}. \quad (\text{C.13})$$

The bare Killing value sits +0.345% above the Standard Model value. The correction is the geometric-dynamics conservation law

$$\frac{1}{\alpha_{\text{SM}}} = \frac{1}{\alpha_W} + \frac{1}{N_c} - \frac{\tau^2\pi}{2}, \quad (\text{C.14})$$

equivalently

$$N_c \left(\frac{1}{\alpha_{\text{SM}}} - \frac{1}{\alpha_W} + \frac{\tau^2\pi}{2} \right) = 1. \quad (\text{C.15})$$

Here α_{SM} is the screened weak coupling returned from the window-capacity coupling α_W and the two correction terms, and $g_2(M_G) = \sqrt{4\pi\alpha_{\text{SM}}}$. The colour-number term is the ratio of the adjoint Casimir to the total fermion Dynkin index,

$$\frac{1}{N_c} = \frac{C_2(\text{SU}(2))}{\sum_f T(R_f)} = \frac{2}{6} = \frac{1}{3}, \quad (\text{C.16})$$

with $C_2(\text{SU}(2)) = 2$ and $\sum_f T(R_f) = 6$ the Dynkin sum of the three generations of doublets, so the screening is the content ratio one over the colour number. The torsion term $-\tau^2\pi/2$ is the Einstein-Cartan correction of the geometric dynamics, the square of the torsion modulus $\tau^2 = 1/2500$ times the half-period $\pi/2$, the Euclidean causal-horizon factor $T_{\text{eff}} = k/(2\pi)$ of the window. The torsion enters the couplings only through its square, because the totally antisymmetric torsion contributes to the kinetic term of the geometric dynamics at order τ^2 , so the correction is the torsion content ratio times the period factor. The bare coupling α_W is the unscreened Killing value, and the physical coupling α_{SM} is the screened value, which is why the physical coupling, not the bare one, stands on the left of the law. The colour-number term $1/N_c$ and the torsion term $\tau^2\pi/2$ remove the overshoot of the bare Killing value, and the resulting coupling is

$$g_2 = \sqrt{\frac{4\pi}{1/\alpha_W + 1/N_c - \tau^2\pi/2}} = 0.508848. \quad (\text{C.17})$$

The colour term is the Casimir-over-Dynkin ratio $C_2(\text{SU}(2))/\sum_f T(R_f) = 1/N_c$ established above. The torsion term combines the quadratic torsion contribution, the Euclidean period, and the conformal normalisation:

$$\frac{\tau^2\pi}{2} = (2\tau^2)(2\pi)\xi = \frac{4\pi\tau^2}{N_g}, \quad (\text{C.18})$$

each traced to the spectrum. The factor $2\tau^2$ is the second-order torsion term of the Einstein-Cartan curvature $\lambda_{\text{EC}} = N_g(1 + \tau/2)^2 + 6 = 14 + 8\tau + 2\tau^2$ of Section C.2. The totally antisymmetric torsion $T^a_{bc} = (\tau/L)\varepsilon^a_{bc}$ contributes the six-component square $2\tau^2$. The factor 2π is the Euclidean period of the spectral sum. The factor $\xi = (d-2)/(4(d-1)) = 1/8 = 1/N_g$ is the conformal scalar-curvature coupling whose identity $N_g\xi = 1$ closes the law. The torsion contribution is therefore $2\tau^2 \cdot 2\pi$, normalised by the generator count through $\xi = 1/N_g$. Both terms of the conservation law are proportional to ξ : $1/N_c = N_g\xi/N_c$ and $\tau^2\pi/2 = 4\pi\tau^2\xi$. The hypercharge coupling follows from the same geometry through the squash normalisation $g_1 = g_2\kappa(2\tau)$, with $\kappa^2(s) = (1+s)/(1-2s)^{5/2}$ at $s = 2\tau$. The isometry breaking $\text{SU}(2)_R \rightarrow \text{U}(1)_R$ squashes the internal S^3 along the σ_3 axis, the $\text{U}(1)_Y$ direction, giving the metric

$$ds^2 = (1+s)\sigma_1^2 + (1+s)\sigma_2^2 + (1-2s)\sigma_3^2, \quad (\text{C.19})$$

with eigenvalues $\lambda_1 = \lambda_2 = 1+s$ and $\lambda_3 = 1-2s$. The volume ratio is $V_3(s)/V_3(0) = \sqrt{\det g} = (1+s)\sqrt{1-2s}$, and the σ_3 -axis inverse metric is $g^{33} = 1/(1-2s)$. The $\text{U}(1)_Y$ normalisation is the volume ratio times the third moment of the axis inverse metric,

$$\kappa^2(s) = \frac{V_3(s)}{V_3(0)} (g^{33})^3 = (1+s)\sqrt{1-2s} (1-2s)^{-3} = \frac{1+s}{(1-2s)^{5/2}}. \quad (\text{C.20})$$

The third moment is the Killing normalisation of the $\text{U}(1)_Y$ gauge field on the squashed internal space. The F^2 contraction of the σ_3 -axis zero mode carries three inverse-metric factors, two from the field strength and one from the polarisation, so the normalisation integral is $\int \sqrt{g} (g^{33})^3 d^3x$ normalised to its round value. At $s = 0$ (round S^3) $\kappa^2 = 1$, so $g_1 = g_2$, the unification. At $s = 2\tau$ the mixing is $\kappa(2\tau) = 1.13183$. The mixing coefficient is corrected by the content ratio

$$\delta_{g_1} = -\tau r_{23} \sum Y^2 \Delta_f \xi, \quad (\text{C.21})$$

with $r_{23} = 3/(10\sqrt{3})$ the hypercharge-trace hierarchy ratio and the product $\sum Y^2 \Delta_f \xi = (10/3)(3/2)(1/8) = 5/8$ the hypercharge capacity times the fermion conformal weight times the conformal coupling, the same conformal-gauge content that closes the $1/N_c$ term above. The colour coupling carries the long-root bifurcation $g_3 = g_2(1 + \alpha_{\text{GUT}}^2/K)$ with $K = J(J+2)/d = 8/3$, where $J = 2$ is the squash angular momentum and $d = 3$ the internal dimension, the kinetic eigenvalue $J(J+2) = 8$ of the $J = 2$ squash over the internal dimension, the long-root condensation diluting the two-loop self-coupling of the colour factor.

C.5 Two-loop gauge coefficients

The gauge beta-function coefficients are derived from the group-theoretic content. The five fermion representations are the minimal combination selected in the companion construction. The chiral clause is carried by the two semispinor representations of $\text{Spin}(4)$, with the left-handed $(1/2, 0)$ transforming as an $\text{SU}(2)_L$ doublet. The colour algebra is $\mathfrak{su}(3)$ and the fermion carries its fundamental 3. Colour vectoriality pairs the left-handed

colour representation with the right-handed conjugate. The left doublet has the colour assignments 3 and 1, giving Q_L and L_L ; the charged components have the Yukawa mirrors u_R, d_R, e_R , while ν_L carries the Majorana closure. Thus the content consists of two left doublets and three right singlets, with $N_L = 6 + 2 = 8 = N_g$. The scalar content is the colour-singlet zero mode whose electroweak part is $(1, 2)_{1/2}$, with hypercharge fixed by the same charge convention.

The hypercharge assignment is derived next. Given the fermion representations and the Higgs hypercharge $Y_H = 1/2$, the charge convention $Q = T_3 + Y$ gives the Yukawa relations

$$Y_u = Y_Q + \frac{1}{2}, \quad Y_d = Y_Q - \frac{1}{2}, \quad Y_e = Y_L - \frac{1}{2}. \quad (\text{C.22})$$

The mixed $\text{SU}(2)^2\text{U}(1)$ anomaly vanishes when $3Y_Q + Y_L = 0$, and the gravitational anomaly when $6Y_Q - 3Y_u - 3Y_d + 2Y_L - Y_e = 0$ (the right-handed singlets enter with the opposite chirality sign). Substituting the Yukawa relations into these two equations gives $3Y_Q - 1/2 = 0$, hence $Y_Q = 1/6$ and $Y = (1/6, 2/3, -1/3, -1/2, -1)$. The per-generation hypercharge moment is

$$\sum Y^2 = 6 \cdot \frac{1}{6^2} + 3 \cdot \frac{4}{9} + 3 \cdot \frac{1}{9} + 2 \cdot \frac{1}{4} + 1 \cdot 1 = \frac{10}{3}. \quad (\text{C.23})$$

The one-loop gauge coefficients follow from the group-theoretic content,

$$b_1 = \frac{2}{5}(3 \sum Y^2 + Y_H^2) = \frac{41}{10}, \quad (\text{C.24})$$

$$b_2 = -\frac{11}{3}C_2(\text{SU}(2)) + \frac{2}{3}T_f(\text{SU}(2)) + \frac{1}{3}T_H = -\frac{19}{6}, \quad (\text{C.25})$$

$$b_3 = -\frac{11}{3}C_2(\text{SU}(3)) + \frac{2}{3}T_f(\text{SU}(3)) = -7, \quad (\text{C.26})$$

with $C_2(\text{SU}(2)) = 2$ and $C_2(\text{SU}(3)) = N_c = 3$ the adjoint Casimirs, $T_f(\text{SU}(2)) = T_f(\text{SU}(3)) = 6$ the three-generation Dynkin sums, and $T_H = 1/2$ the Higgs doublet index. The two-loop matrix is derived from the same content. The diagonal entries are

$$B_{ii} = -\frac{34}{3}C_2(G_i)^2 + \left(2C_2^i(F) + \frac{10}{3}C_2(G_i)\right)S_2^i(F) + \left(2C_2^i(S) + \frac{1}{3}C_2(G_i)\right)S_2^i(S), \quad (\text{C.27})$$

and the off-diagonal entries are

$$B_{ij} = 2C_2^j(F)S_2^i(F) + 2C_2^j(S)S_2^i(S), \quad (\text{C.28})$$

with C_2 the Casimir and S_2 the Dynkin index of each representation, summed over the five Weyl states and the Higgs scalar. The Casimir and Dynkin index are defined as follows. The adjoint Casimir of $\text{SU}(N)$ is $C_2(\text{SU}(N)) = N$, and the Dynkin index of a representation R is fixed by $\text{Tr}(T_R^a T_R^b) = S_2(R)\delta^{ab}$. The fundamental doublet of $\text{SU}(2)$ has $C_2 = 3/4$ and $S_2 = 1/2$, and the fundamental triplet of $\text{SU}(3)$ has $C_2 = 4/3$ and $S_2 = 1/2$. For the $\text{U}(1)$ factor, in the GUT normalisation $g_1 = \sqrt{5/3}g_Y$, the Casimir and Dynkin index are $C_2 = (3/5)Y^2$ and $S_2 = (3/5)\sum Y^2$.

The per-generation Weyl content, whose six entries are the Casimirs and Dynkin indices $(C_2^{U(1)}, C_2^{SU(2)}, C_2^{SU(3)}, S_2^{U(1)}, S_2^{SU(2)}, S_2^{SU(3)})$, is

$$\begin{aligned}
Q_L(3, 2)_{1/6} &: \left(\frac{3}{5}, \frac{1}{36}, \frac{3}{4}, \frac{4}{3}, \frac{3}{5}, \frac{1}{6}, \frac{3}{2}, 1 \right), \\
u_R(3, 1)_{2/3} &: \left(\frac{3}{5}, \frac{4}{9}, 0, \frac{4}{3}, \frac{3}{5}, \frac{4}{3}, 0, \frac{1}{2} \right), \\
d_R(3, 1)_{-1/3} &: \left(\frac{3}{5}, \frac{1}{9}, 0, \frac{4}{3}, \frac{3}{5}, \frac{1}{3}, 0, \frac{1}{2} \right), \\
L_L(1, 2)_{-1/2} &: \left(\frac{3}{5}, \frac{1}{4}, \frac{3}{4}, 0, \frac{3}{5}, \frac{1}{2}, \frac{1}{2}, 0 \right), \\
e_R(1, 1)_{-1} &: \left(\frac{3}{5}, 0, 0, \frac{3}{5}, 0, 0 \right),
\end{aligned} \tag{C.29}$$

and the Higgs scalar $(1, 2)_{1/2}$ contributes the real-scalar content $(\frac{3}{5}, \frac{1}{4}, \frac{3}{4}, 0, \frac{3}{5}, 1, 0)$. The diagonal entry B_{11} is evaluated term by term. With $C_2(G_1) = 0$ for the abelian factor,

$$B_{11} = 2 \sum_f C_2^{U(1)}(f) S_2^{U(1)}(f) n_g + 2 C_2^{U(1)}(H) S_2^{U(1)}(H), \tag{C.30}$$

and inserting the hypercharge content, with $\sum_f C_2^{U(1)} S_2^{U(1)} = (3/5)^2 \sum_f Y^4 = (3/5)^2 (\frac{1}{216} + \frac{16}{27} + \frac{1}{27} + \frac{1}{8} + 1)$ and $n_g = 3$, gives $B_{11} = 199/50$. The remaining entries follow by the same insertion. The full matrix is

$$B = \begin{pmatrix} 199/50 & 27/10 & 44/5 \\ 9/10 & 35/6 & 12 \\ 11/10 & 9/2 & -26 \end{pmatrix}. \tag{C.31}$$

Every matrix entry is the quadratic content ratio of the fermion and scalar representations, carried to its Casimir and Dynkin index.

The Yukawa-gauge mixing coefficients A_i are derived from the same content. The gauge beta carries the top-Yukawa term through the invariant $Y_4(F) = (1/d(G_i)) \text{Tr}[C_2^{(i)}(F) Y^a Y^{+a}]$ with $A_i = 2\kappa Y_4/y_t^2$ ($\kappa = 1/2$ for Weyl fermions). The trace is the explicit matrix element

$$\text{Tr}[C_2^{(i)}(F) Y^a Y^{+a}] = 6 y_t^2 [C_2^{(i)}(Q_L) + C_2^{(i)}(u_R)], \tag{C.32}$$

where the factor $6 = 2 \times 3$ is the weak doublet times the colour triplet of Q_L . Each fermion block contributes the same 6. The Yukawa contraction $Q_L^b H^a u_R \varepsilon_{ab}$ closes the SU(2) indices through $\sum_a \varepsilon_{ab} \varepsilon_{ab'} = \delta_{bb'}$, so the u_R trace carries the weak-contraction factor $2 = \dim \text{SU}(2)$. Hence

$$A_i = \frac{6}{d(G_i)} [C_2^{(i)}(Q_L) + C_2^{(i)}(u_R)], \tag{C.33}$$

with $d(G) = (1, 3, 8)$ and the Casimirs $C_2^{(1)}(Q_L) = \frac{3}{5} \frac{1}{36}$, $C_2^{(1)}(u_R) = \frac{3}{5} \frac{4}{9}$, $C_2^{(2)}(Q_L) = \frac{3}{4}$, $C_2^{(2)}(u_R) = 0$, $C_2^{(3)}(Q_L) = C_2^{(3)}(u_R) = \frac{4}{3}$. The three entries are

$$A = \left(\frac{17}{10}, \frac{3}{2}, 2 \right), \tag{C.34}$$

which reproduces the standard two-loop values directly from the representation content.

C.6 The top-Yukawa and Higgs-quartic beta functions

The scalar-sector running uses the top-Yukawa and Higgs-quartic beta functions, evaluated on the same content by the general two-loop formulas of [23] specialised to the standard-model content in [24]. The conventions here are $V = -m^2|H|^2 + \lambda|H|^4$ and the GUT normalisation $g_1 = \sqrt{5/3}g_Y$. [24] writes the potential as $m^2\varphi^\dagger\varphi + (\lambda_{LX}/2)(\varphi^\dagger\varphi)^2$, so the rescaling $\lambda = \lambda_{LX}/2$, $\beta_\lambda = \beta_{\lambda,LX}/2$ (a pure coupling rescaling) converts their results to the convention used here. The content enters through the trace invariants

$$\begin{aligned} Y_2(S) &= \text{Tr}[3H^\dagger H] = N_c y_t^2, & H(S) &= \text{Tr}[3(H^\dagger H)^2] = N_c y_t^4, \\ \chi_4(S) &= \frac{9}{4} \text{Tr}[3(H^\dagger H)^2] = \frac{9}{4} N_c y_t^4, \end{aligned} \quad (\text{C.35})$$

and

$$Y_4(S) = \left(\frac{17}{20} g_1^2 + \frac{9}{4} g_2^2 + 8g_3^2 \right) y_t^2, \quad (\text{C.36})$$

where the gauge coefficients are themselves the one-loop content of Section C.5, the anti-commutator $-3g^2\{C_2(F), Y\}$ evaluated per gauge group,

$$\begin{aligned} \frac{17}{20} &= 3[C_2^{\text{U}(1)}(Q_L) + C_2^{\text{U}(1)}(u_R)] = 3\left(\frac{3}{5}\frac{1}{36} + \frac{3}{5}\frac{4}{9}\right), & \frac{9}{4} &= 3C_2^{\text{SU}(2)}(Q_L) = 3 \cdot \frac{3}{4}, \\ 8 &= 3[C_2^{\text{SU}(3)}(Q_L) + C_2^{\text{SU}(3)}(u_R)] = 3\left(\frac{4}{3} + \frac{4}{3}\right). \end{aligned} \quad (\text{C.37})$$

The family count $n_g = 3$ entering below is the window-capacity generation number of Section 4.1.

The one-loop top Yukawa is

$$\frac{\beta_{y_t}^{(1)}}{y_t} = \left(\frac{3}{2} + N_c \right) y_t^2 - \frac{17}{20} g_1^2 - \frac{9}{4} g_2^2 - 8g_3^2 = \frac{9}{2} y_t^2 - \frac{17}{20} g_1^2 - \frac{9}{4} g_2^2 - 8g_3^2, \quad (\text{C.38})$$

the Yukawa self-coupling $3/2$ plus the colour trace $Y_2(S) = N_c y_t^2$. The U(1) coefficient $17/20$ is in the GUT normalisation, i.e. $-(17/12)g_Y^2$ in the hypercharge coupling. The two-loop top Yukawa is

$$\begin{aligned} \frac{\beta_{y_t}^{(2)}}{y_t} &= -12 y_t^4 + \left(36 g_3^2 + \frac{225}{16} g_2^2 + \frac{393}{80} g_1^2 \right) y_t^2 - 108 g_3^4 - \frac{23}{4} g_2^4 + \frac{1187}{600} g_1^4 \\ &\quad + 9 g_2^2 g_3^2 + \frac{19}{15} g_1^2 g_3^2 - \frac{9}{20} g_1^2 g_2^2 + 6 \lambda^2 - 12 \lambda y_t^2, \end{aligned} \quad (\text{C.39})$$

with the content structure

$$-12 = \frac{3}{2} - \frac{9}{4} N_c - \frac{9}{4} N_c, \quad 36 = \frac{5}{2} \cdot 8 + 16, \quad \frac{225}{16} = \frac{5}{2} \cdot \frac{9}{4} + \frac{135}{16}, \quad \frac{393}{80} = \frac{5}{2} \cdot \frac{17}{20} + \frac{223}{80}, \quad (\text{C.40})$$

$$\frac{1187}{600} = \frac{9}{200} + \frac{29}{45} n_g, \quad -\frac{23}{4} = -\left(\frac{35}{4} - n_g \right), \quad -108 = -\left(\frac{404}{3} - \frac{80}{9} n_g \right), \quad (\text{C.41})$$

where the first line is the χ_4 - and Y_4 -content of the trace invariants above, and the remaining coefficients (16, 135/16, 223/80, 9/200, 29/45, 35/4, 404/3, 80/9, 9, 19/15, 9/20, 6, 12) are the universal two-loop numbers of the general evaluation. The one-loop Higgs quartic is

$$\beta_\lambda^{(1)} = 24 \lambda^2 - 3\lambda \left(3g_2^2 + \frac{3}{5} g_1^2 \right) + \frac{3}{8} \left(2g_2^4 + (g_2^2 + \frac{3}{5} g_1^2)^2 \right) - 6 y_t^4 + 12 \lambda y_t^2, \quad (\text{C.42})$$

whose Yukawa part is the content combination $4Y_2(S)\lambda - 4H(S) = 12\lambda y_t^2 - 6y_t^4$. The full two-loop Higgs quartic is

$$\begin{aligned}\beta_\lambda^{(2)} = & -312\lambda^3 + 36\lambda^2\left(3g_2^2 + \frac{3}{5}g_1^2\right) - 144\lambda^2 y_t^2 - \frac{73}{8}\lambda g_2^4 + \frac{117}{20}\lambda g_2^2 g_1^2 + \frac{1887}{200}\lambda g_1^4 \\ & + \left(\frac{17}{2}g_1^2 + \frac{45}{2}g_2^2 + 80g_3^2\right)\lambda y_t^2 - 3\lambda y_t^4 + 30y_t^6 \\ & - 32g_3^2 y_t^4 - \frac{8}{5}g_1^2 y_t^4 - \frac{9}{4}g_2^4 y_t^2 + \frac{63}{10}g_2^2 g_1^2 y_t^2 - \frac{171}{100}g_1^4 y_t^2 \\ & + \frac{305}{16}g_2^6 - \frac{289}{80}g_2^4 g_1^2 - \frac{1677}{400}g_2^2 g_1^4 - \frac{3411}{2000}g_1^6,\end{aligned}\tag{C.43}$$

with the content structure

$$\begin{aligned}-312 &= -78 \cdot 4, & 108 &= 54 \cdot 2, & -144 &= -24 \cdot Y_2(S)/y_t^2 \cdot 4, \\ \frac{17}{2} &= 10 \cdot \frac{17}{20}, & \frac{45}{2} &= 10 \cdot \frac{9}{4}, & 80 &= 10 \cdot 8,\end{aligned}\tag{C.44}$$

$$\begin{aligned}30 &= \frac{60}{2}, & -32 &= -\frac{64}{2}, & -\frac{8}{5} &= -\frac{16}{5}/2, \\ -\frac{9}{4} &= -\frac{9}{2}/2, & \frac{63}{10} &= \frac{63}{5}/2, & -\frac{171}{100} &= -\frac{171}{50}/2,\end{aligned}\tag{C.45}$$

$$\begin{aligned}\frac{305}{16} &= \left(\frac{497}{8} - 8n_g\right)/2, & -\frac{289}{80} &= -\left(\frac{97}{40} + \frac{8}{5}n_g\right)/2, \\ -\frac{1677}{400} &= -\left(\frac{717}{200} + \frac{8}{5}n_g\right)/2, & -\frac{3411}{2000} &= -\left(\frac{531}{1000} + \frac{24}{25}n_g\right)/2,\end{aligned}\tag{C.46}$$

$$-\frac{73}{8} = -\left(\frac{313}{8} - 10n_g\right), \quad \frac{1887}{200} = \frac{687}{200} + 2n_g,\tag{C.47}$$

where the $\lambda y_t^2 g^2$ terms are the $10 \lambda Y_4(S)$ content, the y_t^6 , $g^2 y_t^4$, $g^4 y_t^2$ terms are the fermionic traces evaluated on the top content, and the n_g -carrying coefficients are the fermionic Dynkin sums over the three generations. Every coefficient above is displayed as a content combination (or a documented universal two-loop number). None is imported from a table, and each rational identity is elementary (the displayed equalities are the decompositions).

C.7 The Landau-Zener ladder

The generation masses are suppressed exponentially by the extrusion of the modes $n = \{0, 2, 4\}$ under the non-adiabatic squeezing of the scale flow. The extrusion is a Landau-Zener two-level crossing in the scale flow $L(k) = C/k$, with the avoided-crossing Hamiltonian

$$H(t) = \begin{pmatrix} vt & \Delta \\ \Delta & -vt \end{pmatrix}, \quad P = e^{-2\pi\Delta^2/v},\tag{C.48}$$

and the LZ index $\gamma = 2\pi\Delta^2/v$ decomposes into the adiabatic window width and the non-adiabatic torsion. The mass ratios are $m_i \propto e^{-\alpha n_i}$, and the sector index is fixed by the

internal ladder. The up-sector index is the adiabatic window width minus the non-adiabatic torsion correction,

$$\alpha_{\text{up}} = kL - 2\tau = 2.4535, \quad (\text{C.49})$$

where kL is the adiabatic part (the extrusion of a mode across the full coarse-graining window) and $2\tau = n_{\text{broken}}\tau$ is the non-adiabatic deficit (the Einstein-Cartan torsion of the two broken generators, each carrying τ), so the top-to-charm ratio is $m_t/m_c = e^{2\alpha_{\text{up}}}$, and the difference $kL - \alpha_{\text{up}} = 2\tau$ is exact. The down and lepton sectors descend by the step

$$\Delta = 6(1 - n_s)kL_{\text{CMB}}, \quad (\text{C.50})$$

the product of the six generators of the $\mathfrak{so}(4)$ isometry, the spectral tilt $1 - n_s$, and the microwave-pivot window kL_{CMB} . The down and lepton steps are split by the 9/8 hypercharge identity

$$\frac{9}{8} = \frac{1}{1 - (Y_d/Y_l)^2}, \quad (\text{C.51})$$

with $Y_d = 1/3$ and $Y_l = 1$, so $(Y_l^2 - Y_d^2)/Y_l^2 = 8/9$, the lepton step carrying 8/9 of the down step. Writing the lepton step as s and the down step as $(9/8)s$, and using that the lepton rung spans two steps, $(9/8)s + s = 2\Delta$, one obtains $s = 16\Delta/17$ and the down step $(9/8)(16\Delta/17) = 18\Delta/17$. The indices are

$$\alpha_{\text{dn}} = \alpha_{\text{up}} - \frac{18}{17}\Delta, \quad \alpha_{\text{lp}} = \alpha_{\text{up}} - 2\Delta. \quad (\text{C.52})$$

The first-generation step of the down sector carries the colour dilution

$$\alpha_{\text{sd}} = \alpha_{\text{dn}} - \frac{kL_{\text{CMB}}}{6}, \quad (\text{C.53})$$

the microwave window divided by the six generators of $\mathfrak{so}(4)$, the colour dilution of the first-generation extrusion. The symmetry form of this step is $\alpha_{\text{sd}} = \Delta_f(1 - s_0/N_R) = (3/2)(1 - 2\tau/7)$, the fermion conformal weight $\Delta_f = d/2 = 3/2$ times the gravity correction $1 - s_0/N_R$ with $s_0/N_R = 1/175$.

The four coefficients share one physical origin, the non-adiabatic extrusion of the generation modes. The up index is the window width with the torsion subtracted, $kL - \alpha_{\text{up}} = 2\tau$ exactly, because the Einstein-Cartan torsion is the non-adiabatic part of the scale flow, and the LZ exponent is the adiabatic window width minus the non-adiabatic torsion. The factor 2 is the number of broken generators of the squash, $n_{\text{broken}} = 2$. The step Δ is the extrusion rate of the sector ladder, the product of the six $\mathfrak{so}(4)$ generators, the spectral tilt $1 - n_s = \tau \cdot 7/4$, and the microwave window kL_{CMB} . The 9/8 split is the hypercharge weight in the covariant derivative on RP^3 shifting the LZ exponent. The down and lepton doublets carry $Y_d = 1/3$ and $Y_l = 1$, so the lepton rung carries 8/9 of the down rung through $9/8 = 1/(1 - (Y_d/Y_l)^2)$. The colour dilution $kL_{\text{CMB}}/6$ distributes the first-generation extrusion over the six generators of the $\mathfrak{so}(4)$ isometry. Thus each coefficient is a content ratio of the same extrusion closure.

C.8 The top-quark anchor

The top Yukawa is the $(0, 0)$ diagonal overlap of the $n = 0$ spinor mode with the scalar condensate. The overlap is normalised to unity,

$$y_0 = 1, \quad (\text{C.54})$$

because the mass operator is the $(0, 0)$ scalar of $\text{SO}(4)$, the torsion singlet $T_{abc}\gamma^{abc}$, and its matrix element is evaluated explicitly on the generation modes. A generation mode n carries the $\text{SO}(4)$ content $(j_L, j_R) = ((n+1)/2, n/2)$, so the top $n = 0$ is $(j_L, j_R) = (1/2, 0)$. The matrix element of the $(0, 0)$ scalar between a mode and itself is the product of the two $\text{SU}(2)$ scalar-channel Clebsch-Gordan coefficients,

$$y_0 = \langle (j_L, j_R) | (0, 0) | (j_L, j_R) \rangle = C(j_L, 0, j_L) C(j_R, 0, j_R), \quad (\text{C.55})$$

where $C(j, 0, j) = 1$ is the $\text{SU}(2)$ Clebsch-Gordan coefficient of the scalar channel. The identity representation couples $j \otimes 0 = j$ with the unique coefficient one. Hence

$$y_0 = C(\tfrac{1}{2}, 0, \tfrac{1}{2}) C(0, 0, 0) = 1 \cdot 1 = 1. \quad (\text{C.56})$$

This is the explicit wave-function matrix element evaluated on the $n = 0$ generation mode. The scalar channel gives the same unit coefficient for every generation mode, $C(j_L, 0, j_L) C(j_R, 0, j_R) = 1$, while the inter-generation hierarchy is carried by the ladder of Section C.7. The geometric renormalisation group preserves this matrix element, and the top mass follows as $m_t = y_0 v / \sqrt{2}$ from the vacuum expectation value of Section 7.3. Writing $n_s^{\text{tilt}} \equiv 1 - n_s$, the bottom absolute mass is the geometric mean of the strange and top masses dressed by the window evolution,

$$\frac{y_b}{y_t} = e^{-(2\alpha_{\text{dn}} - n_s^{\text{tilt}}(kL_{\text{CMB}} + 2\tau))}, \quad (\text{C.57})$$

$$\text{so } m_b^2 = m_s m_t e^{n_s^{\text{tilt}}(kL_{\text{CMB}} + 2\tau)}.$$

C.9 Charged-fermion masses

The electron exponent is the cascade-content number

$$N_e = (d+1) \left(\sum Y^2 \Delta_f \right) = 4 \left(\frac{10}{3} \frac{3}{2} \right) = 20. \quad (\text{C.58})$$

Here $d+1 = 4$ counts the four cascade levels and $\sum Y^2 \Delta_f = 5$ is the hypercharge capacity of one fermion generation. The electron closure assigns one window factor e^{-kL} to each level-content pair, giving $e^{-N_e kL}$.

The electron mass is the Planck-anchored exponential chain with the low-scale squash correction,

$$m_e = M_{\text{P}} e^{-20kL} (1 - s_0 \kappa), \quad (\text{C.59})$$

where the exponent $20 = 4 \times 5$ is the structural counting of the fourfold and fivefold content, and $s_0 \kappa$ is the squash correction of Section C.4. The two factors have independent structural origins. The factor $4 = d+1$ is the internal dimension $d = 3$ plus one, the four levels

of the spectral cascade. The factor $5 = \sum Y^2 \Delta_f$ is the hypercharge capacity $\sum Y^2 = 10/3$ times the fermion conformal weight $\Delta_f = d/2 = 3/2$, which equals the five fermion representations of one generation, Q_L, u_R, d_R, L_L, e_R , the content of Section 4.1. The product, not the sum, appears because each of the four cascade levels acts on the complete content of one generation, the five species, so the exponent is $20 = (d+1)(\sum Y^2 \Delta_f) = 4 \times 5$, a pure content ratio. The cascade form of the electron mass is

$$m_e = y_0 O_e \frac{v_{\text{dil}}}{\sqrt{2}}, \quad O_e = 1 - \frac{1}{1 + kL^2} e^{-1/kL}, \quad (\text{C.60})$$

the universal Yukawa seed $y_0 = 1$ times the scalar-channel overlap O_e of the lowest retained Dirac mode with the dilaton zero mode, times the dilaton vacuum expectation value v_{dil} at the electron scale. Let χ_0 be a normalised lowest Dirac eigenspinor and let $\hat{\varphi}_{\text{dil}}^{(0)}$ be the scalar zero-mode profile normalised to unit amplitude. Their scalar-channel matrix element is

$$\int_{RP^3} d^3y \sqrt{g} \bar{\chi}_0(y) \hat{\varphi}_{\text{dil}}^{(0)}(y) \chi_0(y) = 1, \quad (\text{C.61})$$

by normalisation and the identity scalar Clebsch–Gordan channel. The finite-window profile is specified by the overlap closure $O_e = 1 - \delta(kL)$ with $\delta(kL) = e^{-1/kL}/(1 + kL^2)$.

The electron-scale dilaton normalisation is defined by

$$v_{\text{dil}} = M_P e^{-20kL} \frac{\sqrt{2}(1 - s_0\kappa)}{y_0 O_e}, \quad (\text{C.62})$$

and substitution gives the equivalent cascade parametrisation

$$m_e = y_0 O_e \frac{v_{\text{dil}}}{\sqrt{2}} = M_P e^{-20kL} (1 - s_0\kappa). \quad (\text{C.63})$$

The descent exponent $20 = (d+1)(\sum Y^2 \Delta_f) = 4 \cdot 5$ is the KK cascade. The dilaton (the scalar zero mode) descends through $d+1 = 4$ levels (the internal dimension 3 plus the scale-flow direction), each level acting on the complete one-generation content $\sum Y^2 \Delta_f = (10/3)(3/2) = 5$ species. Each (level $\times$ species) unit crosses the coarse-graining window once and is suppressed by the LZ survival factor e^{-kL} (one window width kL , the same window whose circumference $4\pi kL$ sets the electroweak hierarchy $\ln(M_G/v)$ of Section C.11). The four steps therefore give $(e^{-kL})^{4 \cdot 5} = e^{-20kL}$. The $O(1)$ prefactor $\sqrt{2}(1 - s_0\kappa)/(y_0 O_e)$ carries the normalisation (the Yukawa factor $\sqrt{2}$ and the seed $y_0 = 1$) and the two level-corrections, namely the finite-size back-reaction δ inside O_e and the $J = 2$ squash $s_0\kappa$, both $O(1)$ corrections to the dominant exponential, in the same two-correction pattern as the electroweak hierarchy $\ln(M_G/v) = 4\pi kL - \ln(3\alpha/\pi) + s_0\kappa$. The compression is thus explicit. The $4 \times 5 = 20$ window crossings collapse into the single exponent $20kL$, and the $O(1)$ factor is the residual normalisation plus the two corrections. The muon-to-electron ratio carries the Euclidean period,

$$\frac{m_\mu}{m_e} = e^{2\alpha_{\text{lp}} + \sqrt{2}\pi}, \quad (\text{C.64})$$

the lepton LZ index plus the entropy-minimum distance $2L = \sqrt{2\pi}$.

C.10 Neutrino masses and the PMNS matrix

The heaviest neutrino mass is set by the Weinberg dimension-five operator with the two-period family scale,

$$m_{\nu_3} = \frac{v^2(2\pi)^2}{k_{\text{GUT}}}(1 + s_0\kappa) = 0.0502 \text{ eV}, \quad (\text{C.65})$$

the electroweak scale squared times the Euclidean-period factor divided by the breaking scale, with the squash correction $s_0\kappa$ of Section C.4. Equivalently, the dimension-five operator has the family scale $\Lambda_W = k_{\text{GUT}}/(2\pi)^2$ and unit coefficient, so $m_{\nu_3} = v^2(1 + s_0\kappa)/\Lambda_W$. The lighter masses are fixed by the hypercharge-trace hierarchy ratios,

$$\frac{m_{\nu_1}}{m_{\nu_2}} = \frac{1}{\text{Tr}(Y^2)} = \frac{3}{10}, \quad \frac{m_{\nu_2}}{m_{\nu_3}} = \frac{1}{\sqrt{3} \text{Tr}(Y^2)} = \frac{3}{10\sqrt{3}}, \quad (\text{C.66})$$

so $m_{\nu_1} = m_{\nu_3}r_{12}r_{23}$ with $r_{12} = 3/10$ and $r_{23} = 3/(10\sqrt{3})$. These content ratios define the hypercharge-trace mass texture

$$M_\nu = m_{\nu_3}U \text{diag}(1, r_{23}, r_{12}r_{23})U^\dagger, \quad r_{12} = \frac{1}{\text{Tr}(Y^2)}, \quad r_{23} = \frac{1}{\sqrt{3} \text{Tr}(Y^2)}. \quad (\text{C.67})$$

Here U is the mixing texture specified below. Diagonalisation verifies that the assembled flavour-basis matrix has the prescribed eigenvalues m_{ν_3} , $r_{23}m_{\nu_3}$, and $r_{12}r_{23}m_{\nu_3}$. The mass hierarchy is therefore the hypercharge-trace closure carried by the same quadratic moment that normalises the gauge sector.

The absolute-mass texture gives the raw rest-mass split

$$\Delta m_{21, \text{raw}}^2 = m_{\nu_2}^2 - m_{\nu_1}^2 = 6.874 \times 10^{-5} \text{ eV}^2. \quad (\text{C.68})$$

The solar comparison applies the finite window-tilt propagation closure to the second light state,

$$m_{\nu_2}^{\text{osc}} = m_{\nu_2}(1 + n_s^{\text{tilt}}), \quad n_s^{\text{tilt}} \equiv 1 - n_s = \tau \frac{7}{4}, \quad (\text{C.69})$$

while m_{ν_1} supplies the absolute floor used in the dark-energy and microwave-temperature closures. This gives

$$\Delta m_{21}^2 = (m_{\nu_2}^{\text{osc}})^2 - m_{\nu_1}^2 = 7.412 \times 10^{-5} \text{ eV}^2, \quad \Delta m_{31}^2 = m_{\nu_3}^2 - m_{\nu_1}^2 = 2.511 \times 10^{-3} \text{ eV}^2. \quad (\text{C.70})$$

The tilt factor is the RP^3 Weyl-window ratio closed in Section C.14. It enters the oscillation propagation map, while the absolute mass sum and cosmological floor retain the rest-mass texture above.

The PMNS mixing angles carry the two-period imprint. The solar angle is the neutrino mass ratio,

$$\sin^2 \theta_{12} = \frac{m_{\nu_1}}{m_{\nu_2}} = \frac{3}{10} = 0.30, \quad (\text{C.71})$$

the light-pair relation of the content-ratio texture. Together with the two angular relations below, it specifies the PMNS closure used in the flavour-basis matrix. The reactor angle is the Euclidean-period imprint times the internal-space geometry factor,

$$\sin^2 \theta_{13} = \left(\frac{1}{2\pi}\right)^2 \frac{\sqrt{3}}{2} = 0.02194, \quad (\text{C.72})$$

and the atmospheric angle is

$$\sin^2 \theta_{23} = \frac{1}{2} + \frac{\text{Tr}(T_3^2)}{(2\pi)^2} = 0.5507. \quad (\text{C.73})$$

All three angles are fixed by the content-ratio texture. The solar angle is the hypercharge-trace reciprocal of Section C.10; the atmospheric angle is the maximum-mixing value $1/2$ corrected by $\text{Tr}(T_3^2) = 2$ over the Euclidean period; and the reactor angle is the period imprint $1/(2\pi)^2$ times the quotient-geometry factor $\sqrt{3}/2 = \sin(\pi/3)$. Together these relations specify the near-tribimaximal two-period texture used in the flavour-basis matrix.

C.11 The electroweak order parameter

The isometry breaking is driven by the $J = 2$ squash mode, the order parameter φ of the internal geometry. Its dynamics is the Landau potential on the curvature axis,

$$V(\varphi) = \frac{1}{2}\xi(R - R_c)\varphi^2 + \frac{\lambda}{4}\varphi^4, \quad (\text{C.74})$$

with $\xi = 1/8$ the conformal coupling and $R_c = 6/\pi$ the critical curvature. The effective mass squared is $m^2 = \xi(R - R_c)$, tachyonic for $R < R_c$, so the condensation is triggered by the curvature crossing, not by a free-spectrum instability. The quartic coupling is fixed by the stationarity of the potential at the squash vacuum $\varphi_0 = 2\tau$,

$$\frac{\partial V}{\partial \varphi} = 0 \quad \Rightarrow \quad \lambda = \frac{\xi(R_c - R_{\text{GUT}})}{(2\tau)^2} = 149, \quad (\text{C.75})$$

with $R_{\text{GUT}} = 6/L_{\text{GUT}}^2$ and $L_{\text{GUT}} = \sqrt{3}/\tau$ the isometry-breaking scale. The Einstein-Cartan torsion algebra fixes the three coefficients of the parity-preserving torsion Lagrangian

$$L_{\text{torsion}} = aT^2 + bT^{bac}T_{abc} + c(T^a_{ab})^2. \quad (\text{C.76})$$

The algebraic-torsion condition, the requirement that the torsion field equation be algebraic rather than dynamical, fixes $b = 4a$, the Holst-Immirzi condition, and the trace term fixes $c = -(a + b/3) = -(7/3)a$. The remaining coefficient $a = M_{\text{G}}^3/4$ is fixed by the canonical normalisation of the torsion kinetic term at the emergence scale, and the stationarity $\partial V/\partial \varphi = 0$ at $\varphi_0 = 2\tau$ then selects the quartic coupling. The vacuum expectation value is the emergence scale times the window-squared line,

$$v = M_{\text{G}} \frac{3\alpha}{\pi} e^{-4\pi kL} (1 - s_0\kappa), \quad (\text{C.77})$$

the exponential $e^{-4\pi kL}$ carrying the hierarchy $v/M_{\text{G}} \sim 10^{-16}$, with the squash correction $s_0\kappa$ of Section C.4. The two factors of the line have independent origins. The prefactor $3\alpha/\pi$ is the product of three pieces, the $J = 2$ squash bifurcation $3\alpha/2$ with $\alpha = 1/(16\pi^2)$ the one-loop weight, the Fourier prefactor $1/\pi$ of the window, and the factor 2 from the double crossing, $(3\alpha/2)(1/\pi)(2) = 3\alpha/\pi$. The exponential $e^{-4\pi kL}$ is the Landau-Zener survival of the $J = 2$ mode crossing the coarse-graining window twice, once at creation and once at stabilisation, each crossing suppressing by $e^{-2\pi kL}$, the same exponential that governs the curvature amplitude of Section C.14.

C.12 The Higgs quartic

The Higgs quartic λ_H is the self-coupling of the pseudo-dilaton, the Higgs identified as the pseudo-dilaton of the scale flow. Its closed form is the dilaton quartic reduced by the one-loop factor,

$$\lambda_H = \frac{\lambda_{\text{dil}} + \sigma_{\text{SM}}}{32\pi^2} = \frac{12\pi + 3}{32\pi^2} = 0.129, \quad (\text{C.78})$$

where $\lambda_{\text{dil}} = 3 \times 4\pi = 12\pi$ is the dilaton self-coupling, the trace anomaly coupling to all three generations, three times the single-generation strong-coupling bound 4π of the Nambu–Jona-Lasinio/Bethe-Salpeter normalisation, and $\sigma_{\text{SM}} = 3$ is the Standard-Model loop contribution, one unit per generation. The tree-level Higgs mass is then

$$m_H = \sqrt{2\lambda_H} v = 124.98 \text{ GeV}, \quad (\text{C.79})$$

against the observed 125.2 GeV, a deviation of -0.17% . The trace-anomaly coefficient that feeds the pseudo-dilaton mass is

$$\beta_{\text{eff}} = \frac{3g_2^2 + g_1^2 + 4y_t^2 + 2\lambda_H}{16\pi^2} + \frac{\lambda_{\text{dil}}}{16\pi^2}, \quad (\text{C.80})$$

the pure-loop Standard-Model part plus the dilaton strong-coupling part. This λ_H is the Higgs quartic, distinct from the colour-singlet order parameter $\lambda = 149$ of Section C.11; the two couplings belong to different sectors.

C.13 CKM mixing and CP

The quark-mixing elements follow from the mass ladder through the Gatto relation. The first-generation ratio carries the hypercharge first-generation factor $|Y_d|/|Y_u| = 1/2$,

$$\frac{m_d}{m_s} = e^{-2\alpha_{\text{dn}}} \left(1 + \frac{|Y_d|}{|Y_u|}\right)^2 = e^{-2\alpha_{\text{dn}}} \left(\frac{3}{2}\right)^2 = 0.05015, \quad (\text{C.81})$$

and the Cabibbo element is the Gatto dominant term,

$$|V_{us}| = \sqrt{\frac{m_d}{m_s}} = 0.2239. \quad (\text{C.82})$$

The Gatto ratio here differs from the absolute-mass ladder ratio $m_s/m_d = e^{2\alpha_{\text{sd}}} = 19.62$ of Section C.7 by 1.6%. The two ratios carry the first-generation colour-dilution step $\alpha_{\text{sd}} = \alpha_{\text{dn}} - kL_{\text{CMB}}/6$ in different powers, the absolute ladder carrying $e^{2\alpha_{\text{sd}}}$ and the Gatto element carrying $e^{-2\alpha_{\text{dn}}}(3/2)^2$. They are therefore kept as separate closure relations. The cross-generation elements are

$$|V_{cb}| = \frac{1}{2}e^{-\alpha_{\text{up}}}(1 - s_0\kappa), \quad |V_{ub}| = \frac{1}{2}e^{-2\alpha_{\text{up}}}(1 + \tau\kappa), \quad (\text{C.83})$$

the adjacent-generation element carrying the squash correction $s_0\kappa$ and the cross-generation element the chiral correction $\tau\kappa$. The three-element product is numerically close to the weak cube,

$$|V_{us}||V_{cb}||V_{ub}| \approx \alpha_W(v)^3, \quad (\text{C.84})$$

with $\alpha_W(v) = g_2(v)^2/(4\pi)$ the electroweak-scale weak coupling of the geometric renormalisation group, distinct from the window-capacity coupling $2/kL^5$ of Section 5.1. The product matches the weak cube to -2.45% and is recorded as the corresponding numerical relation between the ladder and the running coupling.

The CKM phase is the colour-number dilution of the left-right content ratio. The left-right ratio is

$$\frac{N_L}{N_R} = \frac{8}{7}, \quad (\text{C.85})$$

eight left-handed doublets against seven right-handed singlets per generation. The lepton PMNS phase carries the undiluted ratio $\delta_{\text{PMNS}} \sim (8/7)\pi$, and the quark phase is diluted by the colour number,

$$\delta_{\text{CKM}} = \frac{8}{7} \frac{\pi}{N_c} = \frac{8\pi}{21} = 68.57^\circ. \quad (\text{C.86})$$

The Jarlskog invariant is the exact formula with the closed elements and the derived phase,

$$J = |V_{us}||V_{cb}||V_{ub}| c_{12}c_{23} \sin \delta = 3.15 \times 10^{-5}. \quad (\text{C.87})$$

The baryon asymmetry is the Sakharov content, written here as a model relation,

$$\eta_B = \frac{J\alpha_W(v)^2}{56} = 6.09 \times 10^{-10}, \quad (\text{C.88})$$

where J is the Jarlskog invariant above and $1/56 = 1/(N_L N_R)$ is the content count. The factor $56 = N_L N_R = 8 \times 7$ decomposes as $1/56 = \xi/n_R$, combining the conformal-gauge identity $\xi = 1/N_g = 1/8$ with the inverse right-handed count $1/n_R = 1/7$. The two explicit factors $\alpha_W(v)^2$ represent the two weak sphaleron vertices, while the Jarlskog invariant carries the weak-cube numerical relation above. Their product is the five-weak-factor model relation used here. The Sakharov out-of-equilibrium factor is the content normalisation $1/(N_L N_R)$.

C.14 Cosmology

The dark-energy density is the lightest-neutrino mass floor,

$$\rho_\Lambda = Y_u m_{\nu_1}^4 (1 - 4s_0\kappa) = \frac{2}{3} m_{\nu_1}^4 (1 - 4s_0\kappa), \quad (\text{C.89})$$

where $Y_u = 2/3$ is the up-quark hypercharge and the lightest neutrino mass is derived from the Weinberg operator

$$m_{\nu_3} = \frac{v^2(2\pi)^2}{k_{\text{GUT}}}(1 + s_0\kappa), \quad m_{\nu_1} = m_{\nu_3} \cdot \frac{m_1}{m_2} \cdot \frac{m_2}{m_3}, \quad (\text{C.90})$$

with the hierarchy ratios $m_1/m_2 = 1/\text{Tr}(Y^2) = 3/10$ and $m_2/m_3 = 1/(\sqrt{3}\text{Tr}(Y^2))$. The weight $(1 - 4s_0\kappa)$ is the dark-energy-weight symmetry correction. The seesaw mass carries $+s_0\kappa$ (the level transfer from the electroweak scale, whose v carries $-s_0\kappa$), so $m_{\nu_1}^4$ carries $+4s_0\kappa$, and the weight cancels it to order $s_0^2\kappa^2$, keeping the dark-energy density conserved under the $J = 2$ squash level transfer. The physical content of this relation is that the

lightest fermion sets the vacuum-energy floor. The neutrino is the lightest fermion of the spectrum, and its mass is the smallest nonzero energy scale the spectrum carries, so the vacuum energy is anchored at that floor rather than at an arbitrary scale. The weight $Y_u = 2/3$ is the up-quark hypercharge, the hypercharge of the neutral seesaw partner of the up quark, so the floor is the lightest-neutrino mass raised to the fourth power, weighted by the hypercharge of the partner it pairs with through the seesaw.

The entropy integral is the logarithm of the ratio of two Gaussian widths, the Planck width M_P^2 and the vacuum floor $\sqrt{\rho_\Lambda}$,

$$\int \gamma_M = \ln \frac{M_P^2 \sqrt{2\pi + r_{23}}}{\sqrt{\rho_\Lambda}} = 139.2537, \quad (\text{C.91})$$

with $r_{23} = 3/(10\sqrt{3})$ the hypercharge-trace hierarchy ratio. The statistical-mechanical origin is the association entropy of two Gaussians. The Planck anchor is a Gaussian of variance M_P^2 and the vacuum floor a Gaussian of variance $\sqrt{\rho_\Lambda}$. A Gaussian of variance σ^2 has differential entropy $H(\sigma) = \frac{1}{2} \ln(2\pi e \sigma^2)$, and the association of the two ends, twice their entropy difference plus the entropy-minimum distance $\ln \sqrt{2\pi + r_{23}}$, gives

$$\int \gamma_M = 2[H(M_P) - H(\sqrt{\rho_\Lambda})] + \ln \sqrt{2\pi + r_{23}} = \ln \frac{M_P^2 \sqrt{2\pi + r_{23}}}{\sqrt{\rho_\Lambda}}. \quad (\text{C.92})$$

The physical content is the Boltzmann entropy of the scale flow. The scale flow carries two Gaussian measures, one at the Planck anchor of width M_P^2 and one at the vacuum floor of width $\sqrt{\rho_\Lambda}$, and the number of available states between the two ends of the emergence window is the ratio of the two widths, $W = M_P^2 \sqrt{2\pi + r_{23}} / \sqrt{\rho_\Lambda}$. The factor $\sqrt{2\pi + r_{23}}$ is the width of the correlation window between the two ends, the Euclidean period 2π dressed by the hypercharge-trace hierarchy ratio r_{23} , so the entropy is the Boltzmann logarithm $S = \ln W$, and W is the window span over which the flow states are counted. The self-similar branch has $\gamma_M = 0$ (no entropy production, the scale-invariant spectrum), and the infrared end is the maximum-entropy state, $S = \ln W = 139.25$, the Jaynes maximum-entropy endpoint of the duality emergence. This single number is the bridge between the ultraviolet Planck anchor and the infrared vacuum floor. The Hubble rate is the infrared endpoint of the emergence window,

$$H_0 = M_P \sqrt{\pi} e^{-\int \gamma_M}. \quad (\text{C.93})$$

The dark-energy fraction is

$$\Omega_\Lambda = \frac{\rho_\Lambda}{3H_0^2 M_P^2} = \frac{2}{3} + \frac{r_{23}}{3\pi} = 0.68504. \quad (\text{C.94})$$

The algebra is exact. With $H_0^2 = M_P^2 \pi e^{-2\int \gamma_M}$ and $e^{-2\int \gamma_M} = \rho_\Lambda / (M_P^4 (2\pi + r_{23}))$, the ratio is $\rho_\Lambda / (3H_0^2 M_P^2) = (2\pi + r_{23}) / (3\pi) = 2/3 + r_{23}/(3\pi)$, the maximum-entropy closure $2/3$ plus the hypercharge-trace correction $r_{23}/(3\pi)$. The endpoint residual is the remaining Hamiltonian fraction,

$$\Omega_\Sigma = 1 - \Omega_\Lambda - \Omega_b = 0.26570. \quad (\text{C.95})$$

In the linear cosmological propagation this residual is used as a conserved cold source, $\Omega_{\text{cdm}} := \Omega_{\Sigma}$. The conventional cold-dark-matter fraction enters the subsequent reference comparison. The microwave temperature is the photon floor set by the lightest neutrino,

$$T_{\text{CMB}}^{\text{raw}} = \frac{m_{\nu_1} r_{12}}{\pi} (1 - s_0 \kappa) (1 - \tau \Delta_s) = 2.7310 \text{ K}, \quad (\text{C.96})$$

with $r_{12} = 3/10$ and $\Delta_s = 1/2$ the scalar conformal weight. The squash correction $(1 - s_0 \kappa)$ is the same level transfer as the electroweak scale. The photon floor carries $-s_0 \kappa$ while the seesaw mass carries $+s_0 \kappa$. The physical content is that the lightest fermion sets the photon temperature, the neutrino mass scaled by the hypercharge ratio $r_{12} = m_1/m_2 = (N_L - N_R)/\sum Y^2$, the pure content ratio of the chiral difference over the hypercharge capacity, the geometric factor $1/\pi$, and the chiral correction $1 - \tau \Delta_s$ with $\Delta_s = (d-2)/2 = 1/2$ the scalar conformal weight, the same Δ_s that enters the proton-mass correction of Section C.15. The redshift equals the spectrum, in the sense that the photon temperature is the lightest neutrino mass scaled by the content ratio and the period factor. The finite endpoint photon zero mode multiplies the raw floor by

$$C_\gamma = 1 - \frac{\tau}{\pi^2}, \quad T_{\text{CMB}} = C_\gamma T_{\text{CMB}}^{\text{raw}} = 2.72547 \text{ K}. \quad (\text{C.97})$$

This correction is used for the CMB-monopole comparison and for the linear CMB comparison. The internal baryon-density closure remains evaluated with the raw photon floor and the raw Sakharov baryon asymmetry,

$$\Omega_b = \frac{\eta_B^{\text{raw}} n_\gamma^{\text{raw}} m_p}{\rho_{\text{crit}}}, \quad n_\gamma^{\text{raw}} = \frac{2\zeta(3)}{\pi^2} (T_{\text{CMB}}^{\text{raw}})^3,$$

so the Ω_b computation uses $T_{\text{CMB}}^{\text{raw}}$ together with the raw Sakharov ratio η_B .

The primordial spectral tilt is the torsion modulus times the RP^3 Weyl window-weight ratio,

$$1 - n_s = \tau \cdot \frac{7}{4} = 0.035, \quad (\text{C.98})$$

where the numerator is the Weyl-law degree-of-freedom count of the internal fields and the denominator is the four-level window normalisation,

$$\frac{7}{4} = \frac{1 + 2 + 1 + 3}{d + 1}, \quad d = 3.$$

The conformal-curvature expression gives the same rational value as a cross-check,

$$\frac{7}{4} = 1 + \xi R_{\text{LC}} L^2 = 1 + \frac{1}{8} 6.$$

Here 1, 2, 1, 3 are the scalar, transverse-vector, spinor-projected and TT Weyl weights. The primordial curvature amplitude is the spin- $\frac{1}{2}$ Gaussian zero-point suppressed by the window period,

$$\Delta_R^2 = \frac{1}{2} \left(\frac{1}{2\pi} \right)^2 e^{-2\pi k L_{\text{CMB}}} (1 - \tau \kappa) = 2.10 \times 10^{-9}, \quad (\text{C.99})$$

with $\Delta_0^2 = (1/2)(1/2\pi)^2$ the spin- $\frac{1}{2}$ zero-point, $e^{-2\pi k L_{\text{CMB}}}$ the window suppression, and $(1 - \tau \kappa)$ the chiral-squash correction, with $k L_{\text{CMB}} = k L (1 - \tau/4)$ the microwave-pivot

window. These factors close the amplitude directly. The tensor-to-scalar ratio is the Euclidean zero-point of the tensor sector, with the corresponding tensor period factor,

$$r = \left(\frac{1}{2\pi}\right)^2 = 0.02533, \quad (\text{C.100})$$

so the tensor amplitude is $\Delta_t^2 = r\Delta_s^2 = 5.32 \times 10^{-11}$. The value satisfies the current combined upper bound $r < 0.036$ [22] and is recorded here and in Section 10.4 as a bound comparison.

C.15 QCD and the positive colour-singlet scalar mass

The QCD sector closes in three steps. First the condensate energy of the long-root is

$$\Delta E = \frac{1}{8}M_G, \quad (\text{C.101})$$

the curvature gap $K = 8/3$ times the conformal coupling $\xi = 1/8$. The generator mass $m_{\text{gen}} = g_2(2\tau)M_G/\sqrt{2}$ sets the running initial condition, and the two-loop QCD running with top matching gives Λ_{QCD} . Section C.5 derives the two-loop coefficients from the Casimir and Dynkin indices of the representation content. The running equation $\beta_g = g^3/(16\pi^2)(b_i + B_{ij}g_j^2/(16\pi^2) - A_i y_t^2/(16\pi^2))$ then propagates the initial condition $g_3(M_G)$ and the geometric Yukawa $y_0 = 1$. The two-loop terms shift Λ_{QCD} at the level of a few percent. Second, the topological gap is the $l = 2$ scalar mode of RP^3 , whose eigenvalue $\lambda_{\text{glue}} = 8/L^2 > 0$ is the lowest colour-singlet level. Third, the glueball tower is the two-gluon bound-state spectrum,

$$\lambda = 2\lambda_{\text{gluon}} + C_2(J) + n(N_g\xi), \quad N_g\xi = 1, \quad (\text{C.102})$$

with $\lambda_{\text{gluon}} = 4/L^2$ the Killing eigenvalue. The lightest glueball is

$$m_G = 8\Lambda_{\text{QCD}}, \quad (\text{C.103})$$

and the tower ratios are $2^{++}/0^{++} = \sqrt{2}$, $0^{++*}/0^{++} = 3/2$, $0^{-+}/0^{++} = \sqrt{17/8}$. The proton mass is the constituent-quark content with the conformal-duality correction,

$$m_p = N_c\Delta_f\left(1 - \frac{1}{N_g^2\Delta_s}\right)\Lambda_{\text{QCD}}\left(1 + \tau\kappa\sum Y^2\Delta_s\right) = \frac{279}{64}\Lambda_{\text{QCD}}\left(1 + \frac{5\tau\kappa}{3}\right), \quad (\text{C.104})$$

where $1/(N_g^2\Delta_s) = 1/32 = \xi/4$ follows from the two identities $N_g\xi = 1$ and $N_g\Delta_s = 2(d-1)$. The coefficient is the product of the three constituent quarks, $N_c = 3$, the fermion conformal weight $\Delta_f = d/2 = 3/2$, and the conformal-duality factor $31/32$, so $(9/2)(31/32) = 279/64$. The scheme bracket $(1 + \tau\kappa\sum Y^2\Delta_s) = (1 + 5\tau\kappa/3)$ is the correction between the constituent-quark scale and the $\overline{\text{MS}}$ scale, with $\sum Y^2\Delta_s = (10/3)(1/2) = 5/3$ the hypercharge capacity times the scalar conformal weight. The string tension and the deconfinement temperature are

$$\sigma = \frac{\lambda_{\text{TT}}}{\pi}\Lambda_{\text{QCD}}^2 = \frac{14}{\pi}\Lambda_{\text{QCD}}^2, \quad T_d = \frac{\lambda_{\text{vector}}}{N_c}\Lambda_{\text{QCD}}(1 - \tau\kappa) = \frac{4}{3}\Lambda_{\text{QCD}}(1 - \tau\kappa), \quad (\text{C.105})$$

with the chiral-squash factor $(1 - \tau\kappa)$ on the deconfinement scale (deconfinement is accompanied by chiral restoration, so the scale inherits the chiral-squash content), and the self-consistent ratio

$$\frac{\sigma}{T_d^2} = \frac{14}{\pi} \frac{9}{16} (1 - \tau\kappa)^{-2} = \frac{126}{16\pi(1 - \tau\kappa)^2} = 2.6242, \quad (\text{C.106})$$

the content ratio $126/(16\pi) = 2.5068$ softened by the factor $(1 - \tau\kappa)^2$. The QCD scale itself carries the correction $(1 - s_0\kappa/N_g)$ for the Yukawa difference between the scale-invariant geometric Yukawa $y_0 = 1$ and the running Standard Model top Yukawa in the two-loop gauge beta function, applied at the extraction point. This factor is part of the displayed scale definition.

C.16 Primordial nucleosynthesis

The six constants of the weak-rate freeze-out follow from the closed content. The Fermi constant is $G_F = 1/(\sqrt{2}v^2)$ from the closed electroweak scale. The axial coupling is the conformal-weight form

$$g_A = \frac{N_g \Delta_s}{\pi} = \frac{2(d-1)}{\pi} = \frac{4}{\pi}. \quad (\text{C.107})$$

This is the conformal-weight axial coupling of the closed weak-rate model: the generator count $N_g = 2(d-1) = 8$ times the scalar conformal weight $\Delta_s = (d-2)/2 = 1/2$, divided by the Euclidean period π . It is the same conformal-gauge content that closes the gauge sector of Section C.4 and enters the internal freeze-out rate. The electromagnetic self-energy is $\Delta_{\text{EM}} = (1 - 1/(2\pi))\alpha_{\text{em}}(0)\Lambda_{\text{QCD}}$, where the low-energy fine-structure constant is itself derived from the closed content. The fine-structure constant at the internal Z mass is $\alpha^{-1}(M_Z) = 128.208$ (Section 5.2, the two-loop geometric renormalisation group, the same running as $\alpha_s(M_Z)$), and the one-loop QED vacuum polarisation of the same content runs it down to zero momentum,

$$\frac{1}{\alpha_{\text{em}}(0)} = \frac{1}{\alpha(M_Z)} + \frac{2}{3\pi} \sum_f Q_f^2 N_c^f \ln \frac{M_Z}{m_f}, \quad (\text{C.108})$$

with the leptons on the internal ladder masses (m_e, m_μ, m_τ) , the c and b quarks on the internal masses, and the light quarks (u, d, s) frozen at the internal confinement scale Λ_{QCD} . These internally fixed masses and $\alpha(M_Z)$ give $\alpha_{\text{em}}^{-1}(0) = 137.049$, compared with the observed 137.036, a deviation of +0.010%. The radiative correction is $\delta_R = 1 + (1 - \tau)/(8\pi)$. The neutron-proton mass difference is the quark mass difference minus the self-energy,

$$\Delta m_{np} = (m_d - m_u) - \Delta_{\text{EM}}, \quad (\text{C.109})$$

with the quark masses from the ladder of Section C.7. The neutron lifetime is

$$\tau_n = \frac{2\pi^3}{G_F^2 |V_{ud}|^2 (1 + 3g_A^2) m_e^5 f \delta_R}, \quad (\text{C.110})$$

with $|V_{ud}| = \sqrt{1 - |V_{us}|^2}$ and f the phase-space integral. The factor $f = \int F(Z, W) W p(W_0 - W)^2 dW$ combines the Fermi Coulomb function $F(Z, W)$ ($Z = 1$ for the proton) with

the weak-decay phase-space weight and is evaluated numerically at full precision, with $W_0 = \Delta m/m_e$. The retained leading-kinematic order receives a standard recoil and weak-magnetism shift of about -0.2% . Its dominant sensitivity is the endpoint weight: the internal Δm_{np} enters $(W_0 - W)^2$ with roughly four-fold amplification. The freeze-out temperature solves $\Gamma_{\text{weak}}(T) = H(T)$, and the helium yield is

$$Y_p = \frac{2(n/p)_{\text{BBN}}}{1 + (n/p)_{\text{BBN}}}, \quad (n/p)_{\text{BBN}} = e^{-\Delta m/T_f} e^{-t/\tau_n} = 0.2514. \quad (\text{C.111})$$

The effective neutrino species is $N_{\text{eff}} = 3 + \sqrt{3}/(2\pi)^2 = 3.0439$. The five nuclear constants are evaluated at leading conformal-weight, Euclidean-period, and torsion-content order. The perturbative contribution is the one-loop QED running inside $\alpha_{\text{em}}(0)$, evaluated on the closed masses and the closed $\alpha(M_Z)$.

C.17 Correction-factor inventory

Several displayed formulas of the preceding subsections carry correction factors of the schematic form $(1 \pm c\tau\kappa)$ or $(1 \pm cs_0\kappa)$, with $s_0 = 2\tau$ the squash amplitude and κ the isometry-breaking normalisation of Section C.4. These factors are of order a few percent, and they are collected here in one place, together with the physical quantity each one modifies. The factors are:

- $(1 - 4s_0\kappa)$, the dark-energy weight of Section C.14. The seesaw mass m_{ν_1} carries $+s_0\kappa$ through the level transfer, so $m_{\nu_1}^4$ carries $+4s_0\kappa$, and the weight cancels it, keeping $\rho_\Lambda = Y_u m_{\nu_1}^4 (1 - 4s_0\kappa)$ conserved under the $J = 2$ squash transfer.
- $(1 - s_0\kappa)$, the squash level-transfer factor on the electroweak scale v of Section C.11 and on the raw microwave floor $T_{\text{CMB}}^{\text{raw}}$ of Section C.14. Both carry $-s_0\kappa$ while the seesaw mass carries $+s_0\kappa$.
- $(1 + s_0\kappa)$, the seesaw level-transfer factor on the heaviest neutrino scale m_{ν_3} of Section C.10.
- $(1 - \tau\kappa)$, the chiral-squash factor on the deconfinement temperature T_d of Section C.15 (deconfinement is accompanied by chiral restoration) and on the curvature amplitude Δ_R^2 of Section C.14.
- $(1 + \tau\kappa \sum Y^2 \Delta_s) = (1 + 5\tau\kappa/3)$, the scheme correction of the proton mass m_p of Section C.15, between the constituent-quark scale and the $\overline{\text{MS}}$ scale, with $\sum Y^2 = 10/3$ the hypercharge capacity and $\Delta_s = 1/2$ the scalar conformal weight.
- $(1 - s_0\kappa/N_g)$, the Yukawa-difference factor applied at the extraction of Λ_{QCD} in Section C.15, between the scale-invariant geometric Yukawa $y_0 = 1$ and the running Standard Model top Yukawa inside the two-loop gauge beta function.
- $C_\gamma = 1 - \tau/\pi^2$, the finite endpoint photon zero-mode factor multiplying $T_{\text{CMB}}^{\text{raw}}$ in Section C.14. This factor enters the microwave-monopole and linear CMB comparisons. The internal Ω_b closure uses $T_{\text{CMB}}^{\text{raw}}$ and the raw Sakharov η_B .

- $f = \int F(Z, W) W p(W_0 - W)^2 dW$, the pure-kinematics phase-space integral in the neutron lifetime of Section C.16.

The first six factors are built from the isometry-breaking normalisation κ and the squash amplitude $s_0 = 2\tau$. Their common data are:

1. the content invariant $\tau = (N_L - N_R)/(N_f \sum Y^2) = 1/50$ of Section C.2;
2. the squash amplitude $s_0 = n_{\text{broken}}\tau = 2\tau$, with two broken $SU(2)_R/U(1)_R$ directions;
3. the Killing normalisation $\kappa^2(s) = [V_3(s)/V_3(0)](g^{33})^3 = (1+s)/(1-2s)^{5/2}$;
4. *Geometric moment* $c_Q = a_Q r_Q$. Each quantity responds to the traceless shear $ds^2 = (1+s)\sigma_1^2 + (1+s)\sigma_2^2 + (1-2s)\sigma_3^2$ by the first-order integral $c_Q = d \ln Q / ds|_{s=0} = 2n_3 - n_{\text{eq}}$ (the weighted count of the squash deformation across the content), factorised as an amplitude fraction a_Q (geometric = 1 coupling to both broken generators $n_{\text{broken}} = 2$, chiral = $1/2$ coupling to the single chiral asymmetry $N_L - N_R = 1$) times a content ratio r_Q .

Each coefficient is then $Q \rightarrow Q(1 + c_Q s_0 \kappa)$ with $s_0 \kappa = N_g \tau \kappa / (d+1)$:

- v, V_{cb} carry $-s_0 \kappa$ ($a_Q = 1, r_Q = -1$). A geometric quantity on the full geometry responds to the full amplitude s_0 with the compression sign $-$. The dilaton-stop lengthens by $+s_0 \kappa$, so $v = M_G e^{-\varphi}$ shrinks by $(1 - s_0 \kappa)$.
- m_{ν_3} carries $+s_0 \kappa$ ($a_Q = 1, r_Q = +1$). The seesaw relation $m_{\nu_3} = v^2/M_R$ ($M_R = k_{\text{GUT}}/(2\pi)^2$) combines with the level-transfer conservation $v m_{\nu_3} = v^3/M_R = \text{const.}$ The shifts $v \rightarrow (1 - s_0 \kappa)$ and $M_R \rightarrow (1 - 3s_0 \kappa)$ force $m_{\nu_3} \rightarrow (1 + s_0 \kappa)$.
- T_d and Δ_R^2 carry $-\tau \kappa$, and V_{ub} carries $+\tau \kappa$ ($a_Q = 1/2, r_Q = \mp 1$). The chiral level couples to the chiral asymmetry $\tau = s_0/2$ (the amplitude fraction $a_Q = (N_L - N_R)/n_{\text{broken}} = 1/2$). The sign is $-$ for chiral restoration (T_d, Δ_R^2) and $+$ for the cross-generation element (V_{ub}).
- m_p carries $+\tau \kappa \sum Y^2 \Delta_s = +5\tau \kappa/3$ ($a_Q = 1/2, r_Q = \sum Y^2 \Delta_s = 5/3$). The constituent-vs-MSBAR scheme couples the chiral asymmetry to the hypercharge capacity $\sum Y^2 = 10/3$ and the scalar conformal weight $\Delta_s = 1/2$.
- α_s carries $-s_0 \kappa / N_g = -s_0 \kappa \xi$ ($a_Q = 1, r_Q = -1/N_g$). The Yukawa difference $y_0 = 1$ vs the running top Yukawa is a conformal effect normalised by the conformal coupling $\xi = 1/N_g$ of the duality $N_g \xi = 1$.

The amplitude fraction a_Q and the power $r_Q = 4$ follow from the broken-generator, chiral-asymmetry, and $m_{\nu_1}^4$ content; the seesaw response has $r_Q = +1$. The constituent and Yukawa sectors use the following effective response maps:

1. *Constituent scheme* ($m_p, r_Q = \sum Y^2 \Delta_s = 5/3$). The constituent-quark action of the χ SB sector,

$$S_q = \int \sqrt{g} d^3x \left[\bar{\psi} (i\gamma^\alpha e_a^\mu D_\mu) \psi + G_s (\bar{\psi}\psi)^2 \right], \quad D_\mu = \partial_\mu + ig_1 Y A_\mu, \quad (\text{C.112})$$

with the $U(1)_Y$ covariant derivative, has the gap equation $\delta\Gamma/\delta\langle\bar{\psi}\psi\rangle = 0$ giving $m_q = -2G_s\langle\bar{\psi}\psi\rangle$. Its first-order squash response is

$$\frac{\delta m_p}{m_p} = \tau \kappa \Delta_s \sum_c c Y^2 = \frac{5\tau\kappa}{3}, \quad \sum_c c Y^2 = 6\left(\frac{1}{6}\right)^2 + 3\left(\frac{2}{3}\right)^2 + 3\left(-\frac{1}{3}\right)^2 + 2\left(-\frac{1}{2}\right)^2 + 1 = \frac{10}{3}, \quad (\text{C.113})$$

the chiral asymmetry τ times the Killing normalisation κ times the scalar conformal weight $\Delta_s = (d-2)/2 = 1/2$ of the condensate times the hypercharge spectral sum $\sum Y^2 = 10/3$ (the first non-zero hypercharge moment).

2. *Yukawa difference* ($\alpha_s, r_Q = -1/N_g = -\xi$). The two-loop gauge beta carries the top-Yukawa term through the invariant $Y_4(F) = (1/d(G_a)) \text{Tr}[C_2^{(a)}(F)Y^a Y^{+a}]$ with $A_i = 2\kappa Y_4/y_t^2$ (Luo–Wang–Xiao, Eq. 31, with $\kappa = 1/2$ for Weyl fermions), the explicit trace $\text{Tr}[C_2(F)YY^+] = 6y_t^2[C_2(Q_L) + C_2(u_R)]$ with $6 = 2 \times 3$ the weak-doublet $\times$ colour-triplet factor. The geometric Yukawa $y_0 = 1$ (the $SO(4)$ diagonal overlap) exceeds the running y_t , and the difference is a conformal effect normalised by the Yamabe coupling

$$\delta \ln \alpha_s = -s_0 \kappa \xi = -\frac{s_0 \kappa}{N_g}, \quad \xi = \frac{d-2}{4(d-1)} = \frac{1}{N_g} = \frac{1}{8}, \quad (\text{C.114})$$

the squash amplitude s_0 times the Killing normalisation κ times the conformal coupling ξ of the duality $N_g \xi = 1$ (the same ξ that closes g_1 's $5/8 = \sum Y^2 \Delta_f \xi$).

These displayed equations specify the sectoral moments used by the correction inventory and by the numerical table.

D Theoretical sensitivity

D.1 Sensitivity of the hierarchy

The central values are fixed within the closure prescription of Section 2.3. This appendix quantifies their response to the quantities that carry the chain. The inputs separate into four layers. The observed anchor G_N has a relative experimental uncertainty of order 2×10^{-5} ; it propagates into dimensionful quantities with elasticity +1, while dimensionless quantities satisfy the M_P -rescaling invariance. The structural numbers 2, π , $3/2$, $1/8$ and the geometric factors are exact. The window-capacity fixed point kL and torsion modulus τ are derived intermediates, and their defining prescriptions carry the theoretical sensitivity quantified below.

Table 10 gives the elasticity matrix $d \ln O / d \ln p$ for the principal outputs and the three inputs $p = kL, \tau, M_P$, evaluated by central finite difference on the closed forms of

Table 10. The elasticity matrix, which gives the percentage change of each output per one-percent change of kL , τ , or M_{P} . The large negative entries in the kL column reflect the exponential hierarchy $e^{-4\pi kL}$, e^{-20kL} , and their powers.

Quantity O	Elasticity $d \ln O / d \ln p$		
	$p = kL$	$p = \tau$	$p = M_{\text{P}}$
M_{G}	−1.0	0.0	+1.0
$g_2(M_{\text{G}})$	−2.5	0.0	0
m_t/m_c	+5.0	−0.1	0
v	−32.3	−0.1	+1.0
m_e	−49.9	−0.1	+1.0
m_{ν_3}	−64.7	−1.1	+1.0
ρ_{Λ}	−261.6	−4.5	+4.0
H_0	−129.7	−2.2	+1.0
Ω_{Λ}	0.0	0.0	0

Table 11. The convention chain, showing how a one-percent shift of each upstream convention propagates into the fixed point kL and then into the hierarchy.

Convention (1% shift)	$\rightarrow kL$	$\rightarrow v$	$\rightarrow m_e$	$\rightarrow \rho_{\Lambda}$
Kernel exponent $(1 - y)^p$, $p = 2$	+0.8%	−26%	−40%	−212%
Threshold 4/27	−1.0%	+32%	+50%	+262%
Window kL	—	−32%	−50%	−262%

Appendix C. The dominant sensitivities are those with respect to kL . The electroweak scale v , the electron mass m_e , the heaviest neutrino mass m_{ν_3} , the dark-energy density ρ_{Λ} and the Hubble rate H_0 all carry large negative elasticities, because they are built from exponentials of kL . The relations are $v \propto e^{-4\pi kL}$, $m_e \propto e^{-20kL}$, $m_{\nu_3} \propto v^2$, $\rho_{\Lambda} \propto m_{\nu_1}^4$, $H_0 \propto m_{\nu_1}^2$. A one-percent shift of kL therefore moves v by −32%, m_e by −50%, and ρ_{Λ} by −262%. The mass ratios carry only order-unity elasticities, and the dark-energy fraction $\Omega_{\Lambda} = 2/3 + r_{23}/(3\pi)$ is a pure content ratio whose elasticity is exactly zero. The hierarchy lives in the exponentials of kL , not in Ω_{Λ} .

The elasticity into kL traces back to the three conventions that fix the fixed point, through the spectral-pole condition $V_3 \Pi^2/(32\pi^2) = 4/27$ of Section C.1. Table 11 records this chain. The kernel exponent is the transverse-traceless kernel $K_{\text{TT}} = (1 - y)^2$ regarded as $p = 2$ of the one-parameter family $(1 - y)^p$. The threshold 4/27 is the maximum of the mass-weighted kernel $y(1 - y)^2$. A one-percent shift of the kernel exponent moves the threshold by $p \ln(p/(p + 1)) = -0.81\%$, the fixed point by +0.81% (the elasticity $d \ln kL / d \ln(4/27)$ is exactly −1), and v by −26%. The three conventions therefore sit at the top of a steep amplification chain.

The three upstream quantities have explicit sources. The one-loop transverse-traceless response prescription uses $K_{\text{TT}} = k^4/(k^2 + m^2)^2 = (1 - y)^2$, with the Ward condition

Table 12. Response of the hierarchy to representative variations of the mass fraction y and of the kernel exponent p . The flatness of the extremum softens only the first link ($y \rightarrow$ threshold). The threshold-to- kL and kL -to-output links remain first-order, so a 10% variation of y moves v by about 23%.

Convention variation	$\rightarrow kL$	$\rightarrow v$	$\rightarrow m_e$	$\rightarrow \rho_\Lambda$
$y \rightarrow 1.10 y$	+0.73%	−23%	−36%	−190%
$y \rightarrow 1.30 y$	+6.1%	−196%	−303%	−1590%
$p \rightarrow 1.01 p$	+0.81%	−26%	−41%	−212%

implemented in the subtracted flat-space amplitude and $K_{\text{TT}}(k^2, 0) = 1$. The threshold $4/27$ is the exact maximum of the mass-weighted density $y(1-y)^2$ at $y = 1/3$ and supplies the endpoint-pole normalisation. The torsion modulus is the content ratio $\tau = (N_L - N_R)/(N_f \sum Y^2) = 1/50$. These definitions fix the central closure and provide the upstream quantities varied in the sensitivity calculation.

The mass-weighted density has derivative $[y(1-y)^2]' = (1-y)(1-3y)$ and second derivative -2 at $y = 1/3$. The endpoints have zero weight, so $4/27$ is its unique interior maximum. The closure uses this maximal spectral weight as the endpoint-pole normalisation. The stationary point balances one mass fraction with the two kinetic factors, $(1/3)(2/3)^2$, and is recorded alongside the content identities $N_L = N_g = 8$ and $\sum Y^2 = 10/3$.

Within the Gaussian maximum-entropy endpoint closure, this unique non-zero stationary value is the normalisation carried by the mass-weighted TT density. A displacement of the threshold therefore replaces the endpoint prescription rather than varying a parameter inside the closed chain. The large downstream elasticities measure the rigidity of that prescription: a counterfactual change at the pole closure is transmitted without an adjustable compensating parameter.

Within the stated one-loop response prescription, the transverse-traceless tensor projection supplies the scalar form factor $K_{\text{TT}} = k^4/(k^2 + m^2)^2$. The squared propagator gives the denominator, and the numerator carries the two momentum factors of the projected response. The mass-fraction variable $y = m^2/(k^2 + m^2)$ then gives $K_{\text{TT}} = (1-y)^2$.

The flatness of the extremum softens only the first link of the chain, and the remainder stays first-order. Since $4/27$ is a stationary point, a shift of the mass fraction by Δy moves the threshold only at second order, where $y(1-y)^2 = 4/27 - (y - 1/3)^2 + \mathcal{O}((y - 1/3)^3)$ locally, the curvature being $|[y(1-y)^2]''_{1/3}| = 2$. But the threshold acts on the fixed point kL at first order (the elasticity $d \ln kL / d \ln(4/27) = -1$), and kL acts on the hierarchy at first order through the exponentials of Table 10. The two links downstream of the threshold are therefore not softened by the flatness of the extremum. Table 12 records the response of the principal outputs to representative variations of the upstream quantities.

The threshold is stationary at $y = 1/3$, so a local displacement of y enters it quadratically. The downstream hierarchy then amplifies a counterfactual change of the upstream prescription through its exponential dependence on kL . Table 12 gives representative nu-

merical responses around the adopted prescription.

The representative variations of Table 12 are supplemented by a counterfactual window comparison. The numerical chain uses the smooth Gaussian heat-flow envelope associated with the maximum-entropy and unbiasedness clauses of the companion construction [1]. A sharp Litim step [25] sharply separates retained and discarded modes and therefore provides a diagnostic of profile dependence. Re-solving the fixed point with that step gives $kL = 1.09$, compared with 2.49 for the Gaussian profile, a spread of -56% . Its capacity is $(1.09)^2 = 1.19 < 1.5$, and the capacity map of Section C.1 consequently contains zero retained spinor levels. This comparison displays the dependence of the generation-capacity result on the adopted heat-flow closure.

D.2 The spectral-scale flow and the renormalisation group

The spectral-scale flow is the evolution of the spectral datum with the coarse-graining scale, parametrised by the trajectory $L(k) = C/k$ of Section C.1. Below the emergence scale M_G , the zero modes and lowest Kaluza–Klein excitations dominate the spectral sums. Appendix C of the companion paper [1] identifies this as the regime shared with the standard formulation. Here the effective couplings obey the two-loop renormalisation-group equations of Section C.5, with coefficients fixed by the same group-theoretic content. The spectral-scale flow supplies the boundary condition at M_G , and these equations propagate it to lower scales. This establishes the equivalence of the two scale descriptions throughout their shared low-energy regime.

E Flavour and CP structures (formal closure)

The relations recorded in this appendix form the flavour extension of the spectral data. The absolute neutrino scale and the two oscillation splittings are Tier B outputs in the classification of Section 10. The PMNS mixing angles, CKM phase, Jarlskog invariant, and baryon asymmetry are downstream Tier D relations.

E.1 Neutrinos

The neutrino sector is closed by the Weinberg dimension-five operator. The heaviest mass is

$$m_{\nu_3} = \frac{v^2(2\pi)^2}{k_{\text{GUT}}}(1 + s_0\kappa) = 0.0502 \text{ eV},$$

the electroweak scale squared times the two-period factor divided by the breaking scale, with the squash correction $s_0\kappa$ of Section C.4. The lighter masses descend by the hypercharge-trace hierarchy ratios

$$\frac{m_{\nu_1}}{m_{\nu_2}} = \frac{1}{\text{Tr}(Y^2)} = \frac{3}{10}, \quad \frac{m_{\nu_2}}{m_{\nu_3}} = \frac{1}{\sqrt{3} \text{Tr}(Y^2)} = \frac{3}{10\sqrt{3}},$$

so that

$$m_{\nu_2} = m_{\nu_3} r_{23} = 0.0087 \text{ eV}, \quad m_{\nu_1} = m_{\nu_3} r_{12}r_{23} = 0.0026 \text{ eV},$$

with $r_{12} = 3/10 = (N_L - N_R)/\sum Y^2$ and $r_{23} = 3/(10\sqrt{3})$. The tower is fixed by the two hypercharge-trace ratios, and its lightest member m_{ν_1} is the seed that sets the dark-energy density of Section 8.1.

Oscillation comparisons use the directly measured mass-squared splittings. The raw light-pair rest split is $m_{\nu_2}^2 - m_{\nu_1}^2 = 6.874 \times 10^{-5} \text{ eV}^2$. The solar propagation channel uses the closed spectral-tilt dressing of the second light state,

$$m_{\nu_2}^{\text{osc}} = m_{\nu_2}(1 + n_s^{\text{tilt}}), \quad n_s^{\text{tilt}} \equiv 1 - n_s = \tau \frac{7}{4},$$

which gives

$$\Delta m_{21}^2 = (m_{\nu_2}^{\text{osc}})^2 - m_{\nu_1}^2 = 7.412 \times 10^{-5} \text{ eV}^2, \quad \Delta m_{31}^2 = m_{\nu_3}^2 - m_{\nu_1}^2 = 2.511 \times 10^{-3} \text{ eV}^2.$$

The tilt dressing belongs to the comparison-layer propagation, while the cosmological closure uses the absolute masses.

The mixing angles, in the near-tribimaximal form, are encoded by the same content ratios:

$$\sin^2 \theta_{12} = \frac{m_{\nu_1}}{m_{\nu_2}}, \quad \sin^2 \theta_{13} = \frac{\sqrt{3}}{2(2\pi)^2}, \quad \sin^2 \theta_{23} = \frac{1}{2} + \frac{\text{Tr}(T_3^2)}{(2\pi)^2}.$$

The solar angle is the light-pair mass ratio, the reactor angle carries the two-period imprint times the internal-space factor $\sqrt{3}/2$, and the atmospheric angle carries the isospin trace over the Euclidean period. Together they define the phenomenological mixing texture used in this appendix.

E.2 CKM and CP

The quark-mixing closure is expressed by the Gatto relation and the left-right content ratio. The cross-generation elements follow from the mass ladder of Appendix C.7:

$$|V_{us}| = \sqrt{\frac{m_d}{m_s}}, \quad |V_{cb}| = \frac{1}{2} e^{-\alpha_{\text{up}}}(1 - s_0 \kappa), \quad |V_{ub}| = \frac{1}{2} e^{-2\alpha_{\text{up}}}(1 + \tau \kappa),$$

the Cabibbo element being the Gatto dominant term and the two cross-generation elements the adjacent- and next-generation ladder suppressions. The CKM phase is the colour-number dilution of the left-right content ratio,

$$\delta_{\text{CKM}} = \frac{8}{7} \frac{\pi}{N_c} = \frac{8\pi}{21},$$

and the Jarlskog invariant is the product

$$J = |V_{us}| |V_{cb}| |V_{ub}| c_{12} c_{23} \sin \delta.$$

The baryon asymmetry closes the same content through the Sakharov count,

$$\eta_B = \frac{J \alpha_W(v)^2}{56},$$

where $\alpha_W(v) = g_2(v)^2/(4\pi)$ is the electroweak-scale weak coupling of the geometric renormalisation group and $1/56 = 1/(N_L N_R)$ is the content count. This one quantity is evaluated, because it is the single member of this block that the baryon fraction of Section 8.4 consumes. With the ladder masses of Appendix C.7 the closure above gives

$$\eta_B = 6.09 \times 10^{-10}.$$

The value η_B is the evaluation of $J \alpha_W(v)^2/56$ on the closure above. The Jarlskog invariant and CKM phase supply the structural inputs of this phenomenological relation.

These expressions complete the formal correspondence between the spectral datum and the flavour sector. The complete derivation of each closed form is recorded in Appendix C, Sections C.10 and C.13.

F Cosmological extension and BBN (formal closure)

The primordial nucleosynthesis calculation recorded here depends on the core outputs of Sections 5, 8, and 9, and on the effective relations of Appendix E through CKM unitarity. The weak-rate freeze-out relations use the effective scale choices derived in Appendix C. They form the structural BBN extension of the core closure and are stated without a decimal evaluation or observational comparison.

F.1 Primordial helium

The weak-rate freeze-out closes through six internal constants, none taken from observation. The axial coupling is the conformal-weight form, a closure relation of the content

$$g_A = \frac{N_g \Delta_s}{\pi} = \frac{4}{\pi},$$

the electromagnetic self-energy is

$$\Delta_{\text{EM}} = \left(1 - \frac{1}{2\pi}\right) \alpha_{\text{em}}(0) \Lambda_{\text{QCD}},$$

where the low-energy fine-structure constant $\alpha_{\text{em}}(0)$ is itself internal: it is obtained by running the closed $\alpha^{-1}(M_Z)$ of Section 5.2 down through the one-loop QED vacuum polarisation of the same content (Appendix C.16). The radiative correction is

$$\delta_R = 1 + \frac{1 - \tau}{8\pi},$$

the neutrino-species correction is

$$\delta_N = \frac{\sqrt{3}}{3(2\pi)^2},$$

and the remaining two inputs are $|V_{ud}| = \sqrt{1 - |V_{us}|^2}$ from the CKM unitarity of Appendix E, and the phase-space integral f . The neutron-proton mass difference is

$$\Delta m_{np} = (m_d - m_u) - \Delta_{\text{EM}},$$

and the neutron lifetime is the closed form

$$\tau_n = \frac{2\pi^3}{G_F^2 |V_{ud}|^2 (1 + 3g_A^2) m_e^5 f \delta_R},$$

with $G_F = 1/(\sqrt{2} v^2)$ from the closed electroweak scale. The freeze-out temperature solves $\Gamma_{\text{weak}}(T_f) = H(T_f)$, and the helium yield and the effective neutrino number are the closed forms

$$Y_p = \frac{2(n/p)_{\text{BBN}}}{1 + (n/p)_{\text{BBN}}}, \quad (n/p)_{\text{BBN}} = e^{-\Delta m/T_f} e^{-t/\tau_n},$$

$$N_{\text{eff}} = 3 + \frac{\sqrt{3}}{(2\pi)^2}.$$

The six constants close through the conformal-weight form, the two-period Euclidean imprint, and the torsion content ratio, with the single perturbative ingredient the one-loop QED running inside $\alpha_{\text{em}}(0)$. The complete derivation of the closed forms is recorded in Appendix C, Section C.16.

G The acceleration scale and the endpoint residual

The maximum-entropy endpoint has two projections in the present construction. Its normal Hamiltonian projection is the conserved residual density Ω_Σ used as the cold source in the linear cosmological propagation of Section 8.5. Its local spatial projection is the acceleration residue derived below. The two are the normal and spatial projections of the same endpoint closure.

G.1 The endpoint acceleration residue

Let $\sigma_C = c/H_0$ be the terminal coarse-graining length. The maximum-entropy window with this second moment is Gaussian,

$$W_{\text{IR}}(\Delta) = \exp\left(-\frac{1}{2}\Delta^2\sigma_C^2\right),$$

and its Euclidean period assigns the endpoint wave number

$$k_{\text{IR}} = \frac{1}{2\pi\sigma_C} = \frac{H_0}{2\pi c}.$$

The corresponding acceleration amplitude is $c^2 k_{\text{IR}}$. For a unit spatial direction n_i , the traceless angular tensor in the three-dimensional TT channel is

$$Q_{ij} = n_i n_j - \frac{1}{3}\delta_{ij}, \quad n_i n_i = 1,$$

and its norm is

$$Q_{ij} Q_{ij} = 1 - \frac{2}{3} + \frac{1}{3} = \frac{2}{3}.$$

With the residue normalisation summed over the two physical TT polarisations, the quadratic angular coefficient is

$$C_{\text{TT},3} = 2Q_{ij}Q_{ij} = \frac{4}{3}.$$

The local endpoint entry and its acceleration amplitude are therefore

$$\Sigma_{\text{IR}} = C_{\text{TT},3}(c^2 k_{\text{IR}})^2 = \frac{4}{3} \left(\frac{cH_0}{2\pi} \right)^2 = a_0^2, \quad a_0 = \frac{cH_0}{2\pi} \sqrt{\frac{4}{3}}. \quad (\text{G.1})$$

This gives $a_0 = 1.204 \times 10^{-10} \text{ m s}^{-2}$, compared with the empirical Milgrom scale $1.2 \times 10^{-10} \text{ m s}^{-2}$, a deviation of $+0.36\%$. Since H_0 is the terminal spectral endpoint of Section 8.2, Eq. (G.1) fixes the endpoint value used by the local branch.

G.2 The endpoint residue and the local response

For one endpoint mode q , write the quadratic form as

$$S_2[q] = \frac{1}{2}(\lambda_{\text{IR}} + \Sigma_{\text{IR}})q^2.$$

The unscreened and endpoint variances are $\langle q^2 \rangle_0 = 1/\lambda_{\text{IR}}$ and $\langle q^2 \rangle_{\text{IR}} = 1/(\lambda_{\text{IR}} + \Sigma_{\text{IR}})$. Their ratio is the endpoint spectral residue

$$Z_{\text{IR}} = \frac{\langle q^2 \rangle_{\text{IR}}}{\langle q^2 \rangle_0} = \frac{\lambda_{\text{IR}}}{\lambda_{\text{IR}} + \Sigma_{\text{IR}}}. \quad (\text{G.2})$$

The quasi-static spatial entry is the squared acceleration,

$$\lambda_{\text{IR}} = |\nabla\Phi|^2, \quad y = \frac{|\nabla\Phi|}{a_0}.$$

Because Z_{IR} is a quadratic spectral weight, the coefficient multiplying one field gradient is its root-mean-square amplitude ratio. Equations (G.1) and (G.2) give

$$\mu(y) = \sqrt{Z_{\text{IR}}} = \sqrt{\frac{|\nabla\Phi|^2}{|\nabla\Phi|^2 + a_0^2}} = \frac{y}{\sqrt{1 + y^2}}. \quad (\text{G.3})$$

The same square root follows from the three-dimensional deep-infrared scaling. For $y \ll 1$, the flux is $|\nabla\Phi|\nabla\Phi/a_0$, and the associated local functional $\int d^3x |\nabla\Phi|^3/a_0$ is invariant under $x \mapsto sx$, $\Phi \mapsto \Phi$. Identifying the response directly with the quadratic residue would instead give a quartic functional. Thus the Gaussian variance ratio and the three-dimensional scaling select the same response amplitude.

The endpoint flux is

$$\mathcal{J}_i = \mu \left(\frac{|\nabla\Phi|}{a_0} \right) \nabla_i \Phi.$$

Its Gauss law is

$$\nabla \cdot \left[\mu \left(\frac{|\nabla\Phi|}{a_0} \right) \nabla\Phi \right] = 4\pi G \left(\rho_b + \rho_{\Sigma}^{\text{free}} \right). \quad (\text{G.4})$$

All coefficients in Eqs. (G.1)–(G.4) are fixed by the entropy endpoint, the TT multiplicity, and the three-dimensional angular norm.

G.3 The spherical response

For an isolated spherical baryonic mass, define

$$x = \frac{a_N}{a_0}, \quad y = \frac{a}{a_0}, \quad a_N = \frac{GM_b}{r^2}.$$

Integration of Eq. (G.4) outside the source gives

$$x = y\mu(y) = \frac{y^2}{\sqrt{1+y^2}}.$$

The positive solution is

$$y^2 = \frac{x^2 + x\sqrt{x^2 + 4}}{2}, \quad \frac{a}{a_N} = \left[\frac{1 + \sqrt{1 + 4/x^2}}{2} \right]^{1/2}.$$

For $x \gg 1$ it approaches $a/a_N = 1 + 1/(2x^2) + \mathcal{O}(x^{-4})$. In the isolated deep-infrared limit $x \ll 1$,

$$a(r) = \frac{\sqrt{GM_b a_0}}{r}, \quad v^4 = GM_b a_0.$$

The latter is the fixed baryonic mass-asymptotic-velocity relation of the local endpoint branch.

G.4 The endpoint-source decomposition

Equation (G.4) can be written in Newtonian-source form by defining

$$\rho_{\text{dark,eff}} = \rho_{\Sigma}^{\text{free}} + \rho_{\text{pol}}, \quad \rho_{\text{pol}} = \frac{1}{4\pi G} \nabla \cdot [(1 - \mu) \nabla \Phi].$$

It follows identically that

$$\nabla^2 \Phi = 4\pi G \left(\rho_b + \rho_{\Sigma}^{\text{free}} + \rho_{\text{pol}} \right).$$

The linear cosmological propagation uses ρ_{Σ} as the conserved cold source. In an isolated low-acceleration galaxy the polarisation term supplies the local response, whereas in clusters and mergers the free residual component can separate from collisional baryonic gas and contribute to the metric source. The DESI and SPARC results reported here compare, respectively, the linear cold-source projection and the fixed quasi-static response derived above.

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Declarations

Conflict of interest. The author declares no competing interests.

Data and code availability. No new observational dataset is introduced in this work. The numerical outputs, comparison tables, reproduction scripts, and formal verification files are available in the software archived at Zenodo (DOI <https://doi.org/10.5281/zenodo.22067006>) [26]. This article is archived at Zenodo with DOI <https://doi.org/10.5281/zenodo.22067469>. The computed quantities and observational reference values are listed separately. The reproduction procedure and the scope of the formal checks are described in Appendix B.

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